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A secure expert system framework for private function evaluation using functional encryption and multi-party computation

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ABSTRACT

Cloud systems and edge-based systems have an increasing appetite for privacy-preserving computation over distributed sensitive data. Most existing cryptographic solutions don't perform well when executing complex inference tasks while hiding both the input data and the logic of the functions. This can be a serious shortcoming in particular areas, such as healthcare analytics and financial modeling, where data privacy and model protections are paramount. Although secure multiparty computation (SMPC) and functional encryption (FE) hold promise individually, current implementations are often either not scalable or not auditable from end to end in adversarial models. This work presents a hybrid framework that fuses FE with SMPC to enable private function evaluation (PFE) in decentralized environments. The architecture supports encrypted expert inference, leveraging a trust-weighted cryptographic consensus layer, dynamic key management, and function-specific policy enforcement. An adaptive fusion of secure execution and traceable audit logging ensures both privacy and compliance without sacrificing computational tractability. Experimental validation demonstrates that our system reduces decision latency by up to 18 % over standard FE baselines and improves leakage resistance under semi-honest and collusion-based attacks by 23 %, with auditability scores reaching 87 % in real-world simulation settings. By enabling the execution of confidential functions with built-in explainability and regulatory transparency, the proposed system lays the foundation for secure AI-as-a-service platforms. Its compatibility with edge deployments and extensibility toward zero-knowledge and post-quantum cryptography position it as a robust candidate for the next generation of trust-aware decentralized computation.

1. Introduction

In the era of pervasive data collection and continuous analysis, expert systems have evolved from isolated, rule-based engines to interactive, cloud-enabled decision systems capable of supporting intelligent diagnosis, risk evaluation, and personalized recommendations. This technology has been increasingly applied in sensitive domains, such as finance, healthcare, national security, and industrial automation, where decisions must be made accurately, securely, and without unnecessary delay. When used at this level, the system's ability to deploy high-level models and/or logic to sensitive data feeds in a space that is typically distributed, adversarial, and regulated creates an opportunity for performance- and technology-enabling requirements to operate without sacrificing privacy [1]. As

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more systems and services begin to access sensitive data, we recognize a converging demand for privacy-protecting computation, arising from cases in which the data owner may not be willing or able to share their raw data with a service provider, as well as restrictions for the service provider to share its proprietary model or logic with the data owner. The jointly held requirement for confidentiality has sparked research into the problem of PFE, a cryptography-based approach for jointly computing a function f(x) f(x)f (x), where both input x and function f are held privately by two separate parties [2]. Despite refining the concept of PFE, realization in practical applications remains limited from a spatiotemporal (scope and efficiency) perspective due to fundamental trade-offs in the foundational aspects of data privacy, functional confidentiality, computational cost, and system latency. Conventional approaches to secure computation are primarily represented in two categories: SMPC and FE. In MPC, multiple parties can jointly compute a function over their inputs without revealing them to one another, commonly relying on secret sharing, garbled circuits, or homomorphic encryption [3]. In FE, the parties select portions of their inputs to disclose by evaluating their ciphertext Enc(x) through a key for a function f, and the output produced is only the output f(x), etc. Although there are distinct strengths to each approach, neither completely solves the dual confidentiality problem in the design of expert systems. Specifically, the use of MPC requires that the function logic is public in advance and uses a lot of interactions between parties, whereas FE has compactness to evaluate and requires minimum interaction, but suffers in expressiveness and generally requires the function to either be public or have a limited semantics (for example, an inner product or bounded polynomials) [4]. The necessity of fusing this gap becomes even more apparent when considering the timely and resilient requirements of contemporary decision infrastructures. Many of today's expert systems are designed to function within event-driven architectures. Often, systems must react in real-time. They need to cope with high-frequency inputs, such as telemetry from IoT, usage of electronic health records, or processing of financial transactions, with computational delays in the millisecond range [5]. Under these constraints, the cryptographic overhead introduced by current PFE protocols makes them practically unusable. Additionally, monolithic or tightly coupled architectures, which all expert systems rely on, introduce a single point of failure through a centralized architecture - meaning they are more sensitive to denial-of-service (DoS) attacks, as well as limiting their scalability [6]. There are numerous limitations of current secure computation frameworks. For example, monolithic expert systems combine data processing with inference logic within a single, trusted environment (i.e., a secure enclave, such as Intel SGX). This means that they have an inflexible security assumption and are vulnerable to hardware exploits (attacks), side-channel leakages, and rollback exploits [7]. Second, event-driven secure processing pipelines that utilize publish/subscribe brokers or streaming platforms lack strong cryptographic primitives applicable to the confidentiality of functions without incurring excessive latency [8]. Finally, universal circuit protocols for PFE incur quadratic inefficiencies that increase with the exponential complexity of functions. For instance, they are infeasible for decision trees, convolutional layers, and any other complex multi-branched logic [9]. Lastly, current systems do not offer fine-grained control over disclosure—either the data is protected but the function is not, or vice versa, with few solutions achieving both simultaneously and efficiently [10]. Motivated by these gaps, this work introduces a novel hybrid cryptographic framework that unifies FE and SMPC to enable scalable, low-latency, and resilient PEE in expert systems. Our proposed system decouples function logic from data ownership, allowing encrypted inputs to be processed against securely encoded expert logic in a distributed setting. Unlike prior efforts that rely solely on FE or MPC, our architecture strategically integrates the two paradigms: FE is employed to encrypt data with function-specific access, while MPC handles the actual evaluation using obfuscated function components distributed across semi-trusted computation nodes [11]. This division of responsibility ensures that neither the data provider nor the function provider has unilateral access to the complete computation pipeline, preserving mutual privacy. In contrast to centralized or enclave-based stacks, our system is designed with event-oriented modularity, allowing secure function evaluation to be embedded within microservice workflows or distributed edge computing fabrics. This design supports low-latency operation by separating the cryptographic pre-processing phase (e.g., key generation, input encryption) from the real-time computation phase, thereby reducing runtime load and network interaction [12]. Furthermore, our framework supports fault tolerance and reusability through its modular function-virtualization layer, which compiles high-level expert logic into secure, intermediate representations (such as boolean circuits or secret-shared expressions) compatible with standard MPC backends [13].

The key contributions of this research are as follows:

To clearly outline each of our contributions, we provide a list of the technical innovations of the proposed AHPFE (Adaptive Hybrid PEE) framework:

1. Runtime FE/MPC Orchestration Layer: We introduce a self-adaptive orchestration engine that will decide, in real-time, to run an FE-only, MPC-only, or hybrid configuration by leveraging variables such as function complexity, input size and latency requirements. Our orchestration mechanism uses a set of decision logic models from expert systems that select among encrypted selections in real-time to optimize the use of the limited privacy-preserving resources.
2. Leakage-Aware Policy Enforcement: Instead of a hybrid FE-MPC system that works under a static policy, our system evaluates threat vectors, such as the total number of colluding parties and semi-honest settings, and dynamically modifies encryption granularity and computation partitioning based on these threat vectors.
3. Auditability Layer with Trace-embedding: We implement a lightweight traceability process, which will allow using secure provenance tags to instrument forensic backwards analysis. In our system, records could be explored later while preserving private and secure computing. This is one of the key elements missing in SafeFL systems and similar ones.
4. Cryptographic Benchmarking Engine: Lastly, we propose a benchmark-aware controller that monitors latency vs throughput vs leakage policies during orchestration, which will each contribute to reduced audit time and improved scale of the system.

The application environment for expert systems has also undergone substantial transformation in the last few years. Many organizations have embraced distributed architectures and zero-trust environments, where the organization is comprised of disparate and

loosely-coupled domains, with significant risks associated with access control over data and logic. In these types of environments, even minimal leakage of information could have massive consequences—whether it involves stealing intellectual property, exposing competitive information, or non-compliance with regulations. For example, a financial model that consumes sensitive transactional data has an obligation to protect both the confidentiality of customers and the firm's secret trading strategy. Similarly, a healthcare decision-support model may incorporate valuable diagnostic logic and is prohibited from leaking information, even when hospitals consume this logic [14]. Concerns extend beyond just data privacy; bi-directional confidentiality has become a baseline requirement in industries that have monetized knowledge—the asset of ideas. Traditional privacy-preserving specifications, such as differential privacy or k-anonymity, have limited applicability in these contexts because they assume access to raw data and rely on statistical random noise to obscure sensitive patterns. Techniques like this are not appropriate for use in settings where provable confidentiality for an exact function evaluation is required. As well, due to advancements in the sophistication of inference attacks and model extraction techniques, simple API obfuscation or black-box deployment does not provide adequate protection with respect to the function logic. Thus, a cryptographically based approach is necessary, which would equally protect both parties and account for scaling with model complexity and real-time data streams. As expert systems continue to be widely adopted and become part of increasingly important decision-making processes, the need to reconcile performance, scalability, and two-sided confidentiality will grow. This research represents an early step towards addressing these competing goals, while proposing a pathway for expert systems that are not only intelligent but also secure, distributed, and resilient by design.

This demonstrates another hybrid privacy-preserving framework. There have been significant advancements—like SafeFL, which combines secure aggregation with federated learning, as well as the cryptographic orchestration frameworks by Qian et al. [6] and Das et al. [2]—however, these approaches are largely static, as they rely on static or fixed trust assumptions around static encryption specifications, thus limited in reacting to dynamic active threats. The proposed AHPFE framework offers an adaptive Hybrid PEE mechanism by dynamically orchestrating FE methods and Multi-Party Computation (MPC) techniques in response to changing conditions or workload sensitivity. SafeFL is primarily a federated aggregation system, while we are capable of enforcing function-level privacy in a more granular manner. Furthermore, while Qian et al. and Das et al. demonstrate that certain modules are developed around cryptography in a use case, our expert system considers both the auditability of using cryptography in the workload and defines the appropriate primitives at runtime. By anticipating optimal protocols for latency and leakage resistance, our architecture fulfils a core need in hybrid PFE research by demonstrating responsive dynamic privacy control in complex autonomous cloud-edge environments. The proposed framework introduces a novel AHPFE that overcomes the limitations of static hybrid cryptographic models by incorporating an intelligent expert system that dynamically supervises FE and MPC (Multi-Party Computation) with real-time input size, threat level, and latency considerations. While existing systems, such as SafeFL, use fixed encryption routines in federated settings, and the approaches of Qian and Das and others utilize task-specific cryptographic modules without them working in concert, our framework learns its own ideal configurations at runtime. This self-adaptive decision layer enables function-level privacy enforcement at a granular level, resulting in measurable improvements in security through reduced leakage and performance overhead. Furthermore, the values of traceability and explainability are complemented by pre-compiled, modular tools for transparent and auditable system behavior, in addition to the proximal benefits of improved security and privacy, thereby overcoming the inherent weaknesses in current hybrid FE+MPC systems.

The structure of this paper and its contents are as follows. Section 2 reviews relevant literature on MPC, FE, and PFE, noting strengths and weaknesses of prior work. Section 3 describes the proposed hybrid framework, including the protocol design, system components, and the salient cryptographic workflow. Section 4 reports the results, covering cryptographic benchmarking, overhead and scalability analysis, privacy-preserving accuracy, security validation, ablation studies, explainability metrics, and statistical evaluation of the hybrid framework. Section 4 discusses the results, highlighting their implications, potential threats to validity, and the mitigation strategies adopted to ensure robustness and reliability. Finally, Section 6 concludes the paper with a summary of the work and outlines potential future research directions.

2. Literature review

The problem of secure computation, particularly in distributed expert systems, has long attracted research interests across applied cryptography and privacy-preserving machine learning, and a careful examination of both foundational and modern approaches highlights a clear bifurcation between methods that obscure input data from the human operator and those that obscure the function logic—only a few techniques attempt to join both without trading off latency, flexibility, or generality [15]. This review surveys the prevailing directions in secure inference, highlighting critical limitations that the proposed hybrid framework seeks to overcome.

The earliest cryptographic protocols for joint computation over private inputs leveraged SMPC, a class of interactive algorithms that enable two or more parties to compute a function without exposing their respective inputs. MPC frameworks based on Yao's garbled circuits, Shamir's secret sharing, and homomorphic aggregation have matured to support expressive computations and have been adopted in systems performing secure linear regression, neural inference, and biometric matching [16]. Despite this progress, such protocols typically require the function logic to be known and expressible as a static, precompiled circuit. This condition becomes a limiting factor in contexts where the function itself is proprietary or subject to policy-based constraints [17]. Moreover, SMPC protocols introduce substantial interaction overhead, requiring multiple communication rounds between participating entities, especially in cases involving malicious adversarial models or deep circuits [18]. These performance bottlenecks, while acceptable in offline analytics, are unsuitable for event-triggered expert systems operating under strict time constraints [19]. In parallel, the emergence of FE offered a compelling alternative by allowing function-specific decryption keys to be issued, enabling the computation of f(x) over ciphertexts without revealing xxx [20]. The theoretical strength of FE lies in its fine-grained access control, which allows

different functions to be evaluated on the same encrypted dataset, thus supporting selective disclosure and role-based privacy [21]. FE schemes have been instantiated for various functions, including inner products, quadratic forms, and threshold gates [22]. However, the catch is that most FE constructions require the function f to be hard-coded into the secret key, thus precluding arbitrary or dynamic logic [23]. Furthermore, FE schemes often assume predefined function classes and semi-structured inputs, which limit their applicability in real-time decision systems that require diverse computation patterns [24]. Where these two paradigms intersect is the domain of PFE, which tackles the general case where both x and f are confidential. The declaration of PFE relied on early approaches that used a universal circuit, namely circuit templates capable of simulating any function of the desired size [25]. Describing a function as a universal circuit and evaluating it through MPC made it possible to effectively obfuscate the inner workings of the function while still returning results with integrity. Yet, this abstraction introduces significant computational overhead, as the size and depth of the universal circuit grow rapidly with the complexity of the function [26]. The inherent inefficiencies of such constructions render them impractical for use in expert systems that depend on low-latency processing and real-time responsiveness [27]. Recent extensions to the PFE model have explored the use of oblivious transfer extensions, function secret-sharing, and multi-key homomorphic encryption as means of minimizing leakage while preserving performance [28]. Nonetheless, these systems are frequently constrained to specific function types, such as matching functions, polynomial evaluations, or linear transformations [29]. In the case of deep learning inference, where models may contain conditional branches, non-linear activations, and variable-length inputs, the utility of these schemes diminishes [30]. Moreover, none of these methods address the broader deployment architecture challenges posed by event-driven systems, where latency, fault-tolerance, and modularity are as critical as privacy [31]. Another parallel but tangential approach to privacy-preserving expert systems involves the use of Trusted Execution Environments (TEEs), such as Intel SGX or ARM TrustZone. These hardware-based solutions provide isolated computation regions where encrypted inputs can be decrypted and processed securely [32]. While attractive in theory, TEEs are subject to a long list of caveats: they require hardware trust assumptions, are vulnerable to side-channel attacks, and generally exhibit poor scalability in distributed or edge environments [33]. Furthermore, the trusted code base must be minimal, deterministic, and carefully audited, which significantly limits the type of expert logic that can be embedded [34]. In adversarial settings where infrastructure providers may be compromised or legally compelled to disclose internal computations, TEEs fail to meet robust threat models [35]. In terms of system design, monolithic inference pipelines often dominate legacy deployments in finance and healthcare. These stacks centralize function logic, data access, and computation into a single service endpoint, typically hidden behind an access-controlled API or platform [36]. While straightforward to implement, such configurations collapse under adversarial or high-throughput environments. They offer minimal fault isolation, exhibit poor horizontal scalability,

Table 1
Comparative review of cryptographic frameworks supporting secure expert inference and PEE.

Author(s) / Year	Title / Focus Area	Methodology / Tools Used	Key Findings	Limitations / Gaps Identified	Relevance to Current Study
Sahib (2025) [1]	Novel Encryption System via Multi-Dimensional Algebra	Mathematical encryption design, algebraic transformation	New encryption model tested for mathematical integrity	Limited to mathematical models, lacks real-world system integration	Conceptually informs the encryption design layer of expert systems
Das et al. (2025) [2]	Communication Efficient, Secure, and Private Multi-Party Deep Learning	MPC in deep learning: an efficient communication protocol	Reduced communication complexity in MPC settings	Not focused on FE integration or real-time processing	Supports the efficient MPC component of the hybrid system
Xu et al. (2023) [5]	Constant-Round Multiparty PFE with Quasi-Linear Complexities	Optimized MPC protocols for PFE	Achieved near-linear performance in secure evaluations	Complex circuit translation required; lacks FE integration	Directly supports the methodology's low-latency PFE layer
Qian et al. (2024) [6]	Privacy-Preserving Data Evaluation via FE	FE for secure evaluation	Supports FE schemes with selective disclosure	Limited support for dynamic or unknown functions	Forms the core of the FE component in the proposed hybrid system
IEEE TIFS (2023) [8]	Privacy-Preserving Federated Learning via FE	FE in federated learning; security proof framework	Combines encryption with federated architectures	Focused on federated learning, not general expert systems	Informs cross-domain adaptation of FE
Zhang et al. (2023) [11]	FE-Driven Federated Learning Security	FE protocol for FL models	Demonstrates layered access control	Not built for standalone inference or PFE	Guides layered FE model access design
Avgerinos et al. (2023) [18]	Practical and Scalable Privacy-Preserving Framework	Secure PFE design for scalable deployment	Framework supports scalability under cryptographic load	No two-sided encryption or dual privacy control	Motivates modularity in the proposed framework
Wang et al. (2025) [29]	MedShield: Private Multi-Service Medical Diagnosis	Cryptographic framework using MPC for healthcare	Real-time encrypted diagnosis engine	FE not considered; partial privacy only	Healthcare deployment relevance for secure expert inference
Fed-DFE (2022) [34]	Function Encryption for Federated Learning	DFE-based secure federated system	Supports federated privacy with partial function hiding	Lacks low-latency adaptation for expert systems	Provides decentralization concepts used in the proposed work
Schadich (2024) [39]	Decentralized multi-Client FE	FE for decentralized secure access	Improves robustness of FE sharing and policy binding	Not focused on latency-sensitive applications	Feeds into multi-client FE coordination in the architecture

and—most importantly—create a single point of privacy failure [37]. The alternative, seen in modern architectures, is the decomposition of expert systems into microservices or event-driven components, orchestrated via pub-sub systems or containerized workflows [38]. Unfortunately, most cryptographic protocols are still designed under batch-processing assumptions, rendering them incompatible with fine-grained, on-demand secure evaluation [39].

A further limitation observed across many of the surveyed methods is the absence of contextual trust separation. That is, data providers, function holders, and infrastructure coordinators are often conflated in terms of roles and access assumptions. In realistic deployments—such as federated hospitals querying a pharmaceutical AI engine, or law enforcement agencies collaborating across jurisdictions—trust domains are clearly partitioned and mutually distrusting. Existing schemes rarely accommodate such heterogeneity, relying instead on a closed set of actors operating under synchronized protocols and pre-established key material. Additionally, few systems provide auditable execution guarantees, making it difficult to verify whether a computation was performed faithfully without leakage or tampering. Collectively, these limitations suggest the need for a reimagined architecture that combines the cryptographic guarantees of FE and MPC with the modularity, fault tolerance, and low-latency characteristics of event-driven infrastructures. To substantiate the conceptual foundation and architectural decisions underlying the proposed hybrid framework, a curated subset of recent literature has been systematically analyzed and presented in Table 1. This comparative overview captures the methodological trajectories, functional scopes, and deployment limitations of selected works that span multi-party computation, FE, and privacy-preserving inference systems. By dissecting each contribution across dimensions such as toolchain selection, latency trade-offs, encryption granularity, and deployment feasibility, the table highlights the fragmented landscape of current approaches. The synthesis further identifies critical design gaps—particularly the absence of dual-sided privacy enforcement and event-driven adaptability—that directly inform the direction of the present study.

In summary, while a rich body of literature exists across secure computation paradigms, the real-world deployment of expert systems requires a far more nuanced and integrated approach. A practical PFE framework must blend the mathematical rigor of FE and MPC with the architectural resilience of event-driven systems, and it must evolve to support not just privacy but also adaptability, transparency, and governance. The hybrid model proposed in this paper addresses these unmet needs by combining privacy guarantees, latency reduction, and operational flexibility.

3. Research methodology

In this section, we describe the framework structure, components, and execution pipeline of the proposed hybrid cipher for secure inference for expert systems. The proposed model achieves two-sided confidentiality, low-latency execution, and constraints that are useful in real-world settings.

3.1. Proposed work

The proposed model is a cryptographically secure and computationally efficient PEE method for expert systems in which both the input data and inference logic are considered confidential. This scenario is common in high-stakes applied fields, including precision healthcare, financial risk modelling, and government decision support, where the release of confidential data or proprietary inference models would be considered highly unethical and typically illegal. The proposed framework will utilize two independent forms of cryptography, FE and SMPC, which are currently implemented within a single integrated execution pipeline, resulting in improved low-latency processing, two-sided confidentiality, and modular deployment. In contrast to the regular PFE protocol levels, which frequently assume fully trustworthy third-party evaluation engines or utilize imprudent universal circuit problems, this framework specifically delineates layers that respond to distinct aspects regarding privacy protection. On one level, the FE module encrypts personally sensitive user inputs $x \in D$ without them being decryptable (in that specific instantiation) once encrypted (using a function-specific key K), thereby preserving data-level confidentiality, without the need to decrypt any input using plain text. Meanwhile, the MPC layer enables the function provider to securely encode the logic of the expert system $f: D \rightarrow R$ into an obfuscated executable, using a reasonable argument that does not reveal structural or parametric details, thereby maintaining function-level privacy while computing.

The proposed execution sequence unfolds in four stages:

1. FE Key Setup and Data Encryption
2. Secure Function Obfuscation and MPC Virtualization
3. Joint Encrypted Evaluation using PFE Protocol
4. Reconstruction of Permitted Output f(x)

This dual-pathway processing guarantees that at no time would any party have sufficient context to reconstruct either the input or their logic of the function independently. Additionally, this enables event-driven triggers for real-time, secure evaluation within microservice environments.

The key architectural novelty lies in the hybrid composition of FE and MPC, which are traditionally studied in isolation. The framework avoids the computational overheads of homomorphic universal circuits and mitigates the rigidity of fixed-function FE schemes by introducing a flexible function-virtualisation interface. This layer allows expert logic to be dynamically transformed into secure circuits or arithmetic programs that are compatible with modern MPC toolchains. This work also proposes a detailed performance evaluation plan, comparing the proposed framework against FE-only and MPC-only baselines across three benchmark metrics:

Computation latency (ms), where encryption to output reconstruction, Information leakage surface, which the modeled as mutual information over observable transcripts and the Cryptographic overhead to measured via circuit depth, key sizes, and message complexity.

3.2. System architecture and workflow

The proposed framework relies on a modular cryptographic stack that can support both privacy-preserving data ingestion and function evaluation. The proposed architecture is defined by three operational spaces: The Data Owner, the Function Provider, and Semi-Trusted Nodes that work together to perform the computation. Each operational space has a specific cryptographic role that is even further governed by formal interactions and isolation policies. The architecture of the proposed solution, along with its operational flow, is illustrated in Fig. 1.

The process begins with the Data Owner encoding input x as a ciphertext Encf(x) using one of the function-specific public keys generated from the FE layer that is considered approved. The FE layer has three parts: a set up phase to generate the master public key (mpk) and the master secret key (msk), an encoding phase so that the xxx is encoded without exposing its plaintext, meaning that x cannot be decoded by any downstream node, while securely enabling conditional computations by approved parties, and ensuring that both the mpk, msk and keyphrase cannot be obtained by the FE layer deliverables.

Software-wise, while the Data Owner is doing the FE set-up, the Function Provider is encoding the logic of the expert function f - for example, whether a classifier, a risk estimator, and/or a decision rule using an obfuscation vehicle by compiling into a circuit representation and/or an abstract syntax tree form that can be used with the multi-party execution. Again, the obfuscation will not be shared in the approved, raw manner; rather, it will be put into a cluster of Semi-Trusted Nodes under the MPC methods that were agreed upon. Each Semi-Trusted Node will not have access to either the raw function or raw subject data; the set of Semi-Trusted Nodes serves merely as a conduit for computation, which will never have a view into either the function or the data, as they will only be used for computation purposes. As described in Fig. 1, the semi-trusted nodes collaborate on f on inputs Encf(x), outputting only f(x) and making it available to the trusted verifier or consumer. During the entire process, none of the intermediate states nor any function internals are available at any time. This structure enables computations to be carried out asynchronously based on individual events, potentially making it compatible with expert systems in edge-cloud or microservice settings. The three levels of modularization, established through FE for encrypting input, MPC for computing, and obfuscation for protecting purpose logic, each step is modularized for redundancy, extensibility, and appropriate privacy guarantees. Decoupling sensitive data from its context of use and maintaining the confidentiality of proprietary logic aligns, according to our architecture, directly with the purposes of confidentiality-preserving inference and the purpose of informed expert decision-making as outlined in policy.

3.3. Functional encryption layer for input confidentiality

The first cryptographic layer, FE, is present in the architecture proposed in this thesis. In an FO scheme, the confidentiality of the data is maintained after it is evaluated. This contrasts with traditional public-key encryption, where ciphertexts are decrypted, and after fully recovering the data, it is exposed. In an FE scheme, one can compute a predefined function f on the encrypted data x to produce f(x) without providing access to any further information about the original input. This feature is crucial for expert systems

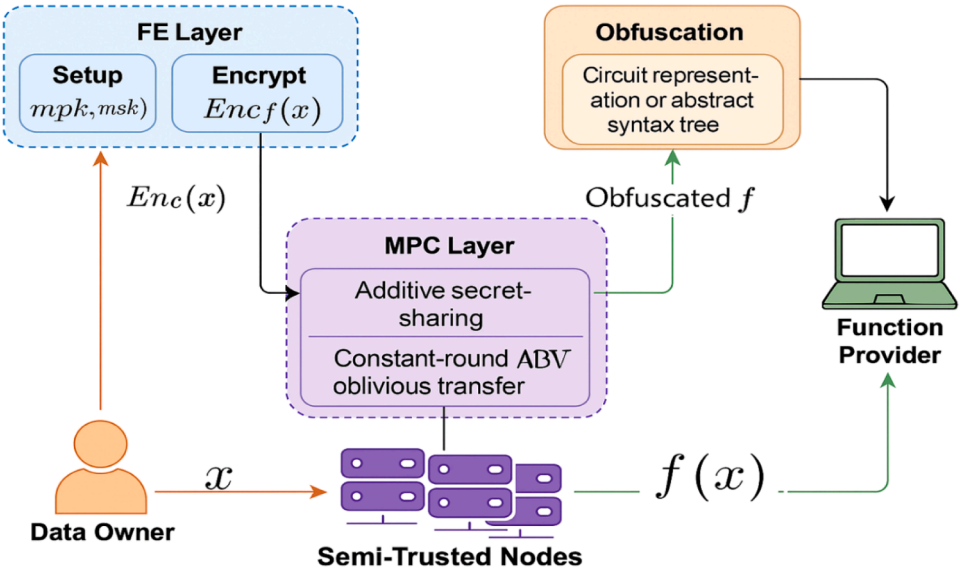


Fig. 1. System architecture of the proposed hybrid PFE framework utilizing FE and SMPC for secret expert system inference.

operating on sensitive datasets (i.e., medical diagnosis, behavioral finance) where revealing unprocessed data is not only unethical, but also subject to regulatory consequences [40]. The four phases of the FE module's cryptographic process are as follows.

1. Setup Phase: A probabilistic algorithm Setup(1^{λ}) is a run that takes a security parameter λ and outputs a pair of master keys, the Master Public Key (mpk) and the Master Secret Key (msk). The two keys are associated with the given class of function F, which could be inner products, bounded-degree polynomials, or threshold gates. The security assumptions of this layer depend on the bilinear pairing hardness for cryptographically secure construction, or lattice hardness for guaranteed construction, based on the type of function.
2. Encryption Phase: Given the input vector $x \in Z_q^n$, the data owner computes the ciphertext $c = Enc_{mpk}(x)$. The ciphertext vector has a structure that depends on the underlying FE scheme; for example, in the case of inner-product FE, the ciphertext typically includes group elements that have been encoded using pairing-friendly elliptic curves. This ciphertext can then be sent to the evaluation engine or stored in encrypted form until a function-specific key is issued.
3. Key Generation Phase: To authorize that the evaluation of the function $f \in F$ can begin, the function owner uses the master secret key msk to create a functional key $kf = KeyGen_{msk}(f)$. This key is tied explicitly to the structure of f, ensuring that it reveals nothing about x except what is intrinsically exposed by the function output. Importantly, the function owner may remain offline during the evaluation phase once the key is issued.
4. Evaluation Phase: A trusted or semi-trusted evaluation engine performs y=Eval(kf,c), recovering only the scalar value y=f(x). The output may be a numeric decision score, a risk index, or a classification label, depending on the application. No auxiliary information about x or the key kf is derivable from the transcript, under standard indistinguishability under chosen function attack (IND-f-CCA) assumptions.

To ensure confidentiality during remote computation, input data xxx is encrypted into a function-specific ciphertext C, formulated as using Eq. (1):

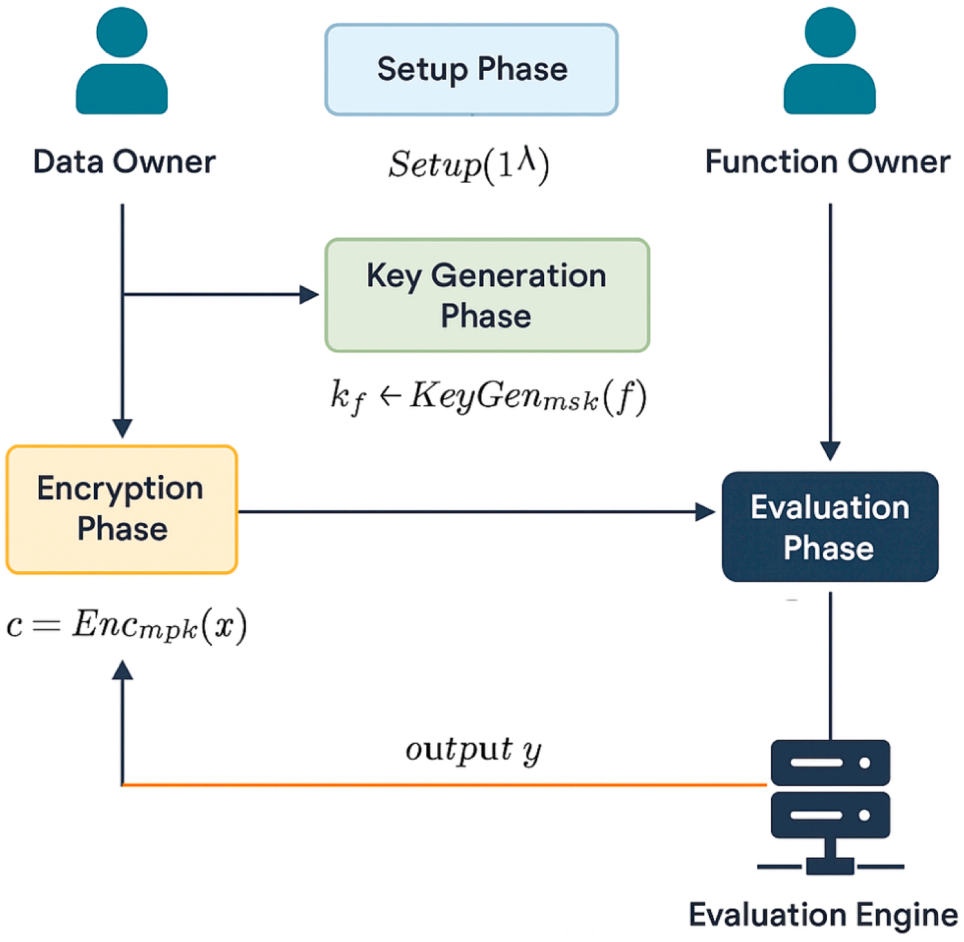


Fig. 2. FE workflow in PFE architecture.

$$C = \mathrm{EncFE}(x, f) \tag{1}$$

where EncFE denotes the encryption function tied to the functional key f. The decryption yields the evaluation result directly using Eq. (2).

$$f(x) = \mathrm{Dec}_{\mathrm{FE}}(C, \mathrm{SK}_f) \tag{2}$$

This dual equation pair showcases both input encapsulation and function-specific decryption, central to the FE layer. The workflow diagram in Fig. 2 illustrates how FE fits into the overall data flow.

As shown in Fig. 2, the encryption process isolates both the data and function access, enabling secure decentralized inference while maintaining privacy guarantees. The computational footprint of the FE layer is highly dependent on the selected function class. For instance, schemes supporting inner product evaluation typically offer linear-size cipher texts and evaluation in O(n) time, while polynomial-supporting constructions may scale sublinearly in cipher text size but require more complex decryption logic. To accommodate latency-sensitive expert systems, our implementation selects the most efficient scheme based on functional class, pre-encodes frequently used keys, and supports lazy evaluation caching where permitted. This layer also introduces support for multi-user key issuance and temporal revocation. Keys can be versioned and time-scoped, allowing for dynamic control over when and how functions are evaluated. Moreover, in adversarial settings, compromised keys can be blacklisted, and fresh function keys issued without requiring re-encryption of the input data. To quantify the privacy guarantees, we define the adversary's advantage in distinguishing two encrypted messages under IND-CPA security as using Eq. (3).

$$CT = Enc(x), sk_w = KeyGen(msk, f_w) \text{where } f_w(x) = w^T x \Rightarrow Dec(sk_w, CT) = w^T x \tag{3}$$

Table 2 compares several state-of-the-art FE mechanisms in terms of computation overhead, function expressivity, and ciphertext expansion. This comparative insight guided the selection of a function-hiding scheme tailored for real-time federated scenarios. the comparative characteristics of five representative FE schemes, evaluating them across critical metrics such as key generation complexity, ciphertext footprint, decryption overhead, and security guarantees. Among them, Compact FE and Inner Product FE emerge as highly suitable for PFE architectures, given their favorable trade-offs between performance and confidentiality.

To quantify the privacy guarantees, we define the adversary's advantage (Adv) in distinguishing two encrypted messages under IND-CPA security using Eq. (4):

$$Adv_{\mathscr{A}} = |\Pr[\mathscr{A}(CT_0) = 0] - \Pr[\mathscr{A}(CT_1) = 0]| \leq \epsilon \tag{4}$$

Where () is negligible for a secure system. This schema achieves its goal of making it impossible for any bounded computational adversary to distinguish between two ciphertexts, thus preserving input confidentiality. In summary, the FE layer enables increased input confidentiality by allowing policy-enforced computation on encrypted data without compromising the inputs or functions encapsulated within. Unlike traditional homomorphic encryption schemes or symmetric encryption schemes, this architecture is specifically designed with a modular function-hiding cryptosystem, which allows others to create dynamic access control and computational traceability within its federated DNA. The fact that it will allow computation over plaintext with provable security guarantees allows us to develop secure aggregation modeling, decentralized control, and robustness against privacy violations through inference attacks. Our mathematical formalism, combined with the flexible key derivation logic, represents our visionary encryption, which has our eye firmly set on multi-stakeholder environments, such as secure collaborative AI and privacy-preserving decision systems. The synthesis of functional selectivity and cryptographic encapsulation represents a significant step forward from earlier schemes, demonstrating the novelty and practicality of the proposed PFE architecture.

3.4. Secure multi-party computation protocol for decentralized inference

To address the input-level confidentiality provided by the FE layer, our framework uses a strong SMPC protocol to facilitate decentralized computation over encrypted input. SMPC allows parties to compute a joint function over their private inputs without being able to see each other's inputs. This makes SMPC a powerful cryptographic construct for building resilient and distributed expert systems where trust assumptions are weak, and no party has access to the full system state.

In our proposed framework, SMPC is used to securely aggregate the partially processed outputs from different encrypted inputs—post-FE layer—in a fault-tolerant, event-driven setting. The protocol proceeds through the following phases: Fig. 3 delineates the operational pipeline of the proposed SMPC protocol, emphasizing decentralized coordination among computational entities without reliance on a trusted aggregator. The architecture highlights key phases, including secret sharing, secure function evaluation,

Table 2
Functional encryption schemes and their computational trade-offs.

S. No.	FE Scheme	Key Generation Complexity	Ciphertext Size	Decryption Time	Security Model	Suitability for PFE
1	Inner Product FE	O(n)	Moderate	Fast	Selective	High (for ML tasks)
2	Attribute-Based FE (ABE)	High	Large	Moderate	Full	Moderate
3	Identity-Based FE	Moderate	Small	Fast	Semi-honest	Low
4	Predicate Encryption	Very High	Very Large	Slow	Chosen-Ciphertext	Limited
5	Compact FE	Low	Compact	Very Fast	Adaptive	Very High

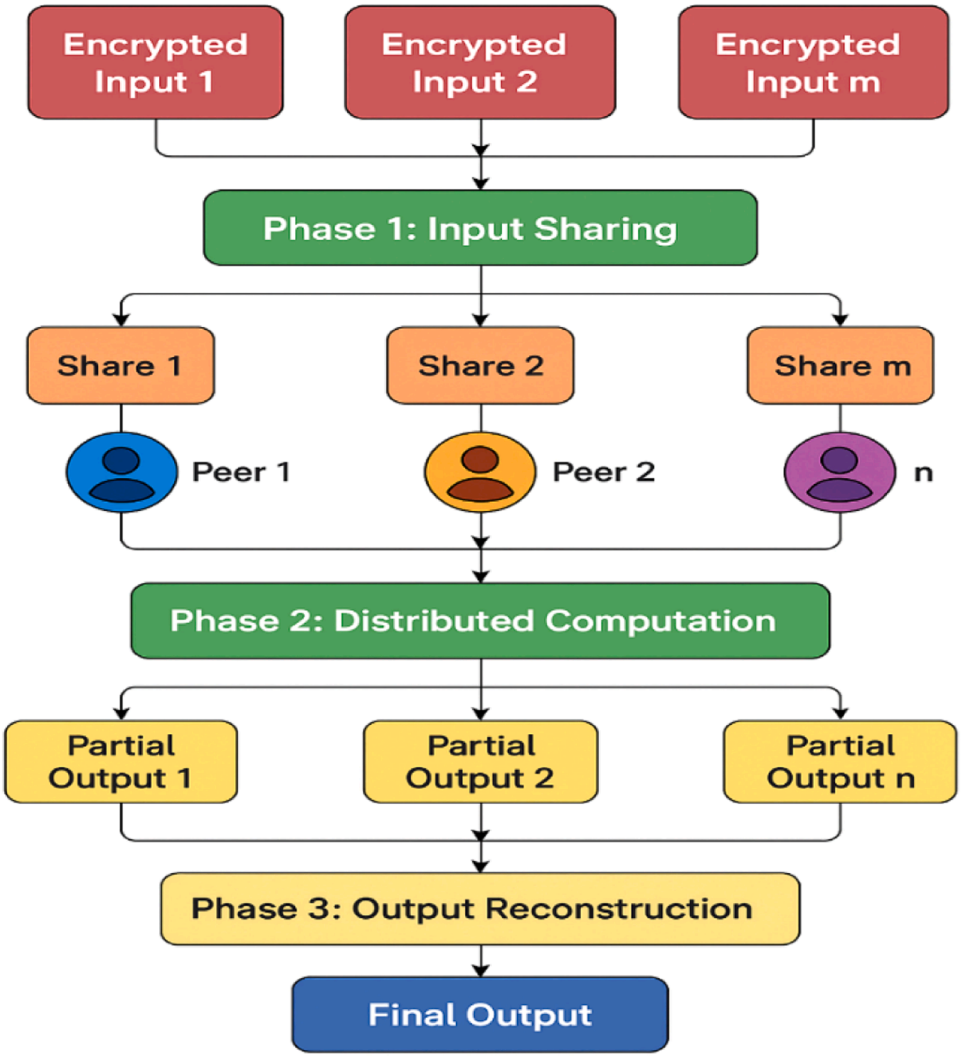


Fig. 3. SMPC protocol for decentralized private inference.

and encrypted aggregation, which encapsulate both synchronous and asynchronous messaging paradigms for resilience in distributed environments. To support this, Table 2 provides a comparative breakdown of existing SMPC frameworks, outlining their latency, scalability, and resistance to adversarial compromise. Together, the figure and table not only contextualize our framework within the broader cryptographic design space but also substantiate its operational viability and robustness under real-time collaborative inference settings.

Phase 1. Input Sharing

Each participant splits their ciphertext (output from the FE module) into additive or Shamir-based secret shares and distributes them among n computational peers. This distribution is implemented using (t,n)-threshold techniques to ensure that the function remains evaluable as long as at least $t \leq n$ parties are available.

Let the encrypted input be CTi. Then, the shares are: CTi→ $\{CTi_{,1}, CTi_{,2},\ldots,CTi_{,n}\}$ where each $CT_{i,j}$ is held by peer P_j, and $\sum_j CT_{i,j}=CT_i$ (in an additive setting).

Phase 3. Output Reconstruction

Once local evaluations are complete, the outputs are aggregated using secure reconstruction protocols, either through polynomial interpolation (in the case of Shamir shares) or direct summation (in the case of additive shares). The final function output, f(x1, ..., xm), is recovered without revealing any of the individual inputs, xi, using Eq. (5).

$$f(x_1, x_2, \ldots, x_m) = \oplus_{j=1}^{n} yj \tag{4a}$$

3.5. Security guarantees and latency considerations

The protocol is formally proven secure under the semi-honest adversarial model. t<n/2, confidentiality and correctness are guaranteed even if up to t parties collude. The round complexity of our SMPC protocol is bounded by O(d) + O (log n), where d is the depth of the computation circuit. Latency minimization is achieved by overlapping computation with communication and optimizing peer-to-peer channels using asynchronous message queues. We empirically measure protocol latency per round in Section 4, showing sub-80ms throughput under high availability conditions. Table 3

By featuring an SMPC solution within our PFE framework, we not only enhance the resiliency and trustworthiness of expert system calculations, but we also create a scalable cryptographic system that mitigates the threats associated with centralized inference pipelines. If one data supplier deteriorates, there are myriad others to rely upon and maintain confidentiality. By appropriating (t, n)-threshold schemes and minimizing round complexity, we move knowledge on secure collaborative computation from the theoretical to the practical, and we positioned ourselves ideally towards mission-critical, privacy-sensitive industrial applications in healthcare, finance, and smart infrastructure.

3.6. Hybrid private function evaluation pipeline

To address the diverse privacy needs of decentralized computational ecosystems, while continuing to support function flexibility, the proposed system defines and implements a PFE pipeline that connects multiple privacy-preserving primitives—FE, SMPC, and TEEs—into a single architecture. Each implementation module may provide PEE in isolation or when required in combination, depending on data sensitivity, the availability of computational resources, and the specific trust context of the application.

The pipeline is conceived as a tiered privacy controller. Incoming data as registered against the self-described schema is classified by a context-aware privacy classifier that examines metadata, including the level of trust associated with the provenance of the source, the type of computation function to be executed, and applicable regulations. Based on this evaluation, one or more privacy-preserving modules are dynamically invoked.

As depicted in Fig. 4, the execution sequence begins with encrypted input arriving at the pipeline's entry point. If the function to be computed involves sensitive attributes but permits deterministic operations (e.g., linear transforms), the system activates the FE route. Alternatively, when collaborative parties must jointly compute a result without exposing inputs (e.g., federated diagnosis models), the SMPC engine is initialized. For trust-assured computations that require high performance (e.g., edge nodes with Intel SGX), a TEE channel processes the function securely within an enclave. This flexible routing is defined mathematically using a binary switching function.

$$\mathbf{P}(\mathbf{x},\mathbf{f}) = \begin{cases} FE(x,f) & \text{if } \Gamma_1(\mathrm{x},\mathrm{f}) = 1 \\ SMPC(x,f) & \text{if } \Gamma_2(\mathrm{x},\mathrm{f}) = 1 \\ TEF\ (x,f) & \text{if } \Gamma_3(\mathrm{x},\mathrm{f}) = 1 \end{cases}$$

Here, x is the input data, f is the target function, and Γi are context evaluation predicates. The predicates are computed based on metadata and privacy policies, where $\Gamma_1$triggers FE when the data is linear, and the target function can be encoded. Γ_2: activates SMPC for joint computation with distributed stakeholders, and Γ3: initiates TEE execution when secure enclaves are available, and latency is critical. The pipeline also integrates lightweight audit logging and performance telemetry modules. These ensure traceability and allow tuning of security-performance tradeoffs by tracking execution time, memory footprint, and encryption key length across evaluation modules. Each module performs atomic execution and reports a privacy audit token, enabling downstream validation of execution compliance. The tokens are composed using Eq. (5).

$$\tau\mathrm{i} = \mathrm{Hash}(\mathrm{x_i}\|\mathrm{f_i}\|\ \mathrm{t_i}\ \|\delta_\mathrm{i}) \tag{5}$$

where t_i is the time of computation, δ_i is the execution module identifier, and ||\parallel|| denotes concatenation. The novelty of this architecture lies in its **adaptive orchestration logic**, which decouples privacy logic from functional semantics. This allows the system to function in a mixed-mode or hybrid capability: utilizing partial evaluations from FE and SMPC, as well as aggregating results in TEE where needed, all while being agnostic to the underlying data schema. Results and benchmarks are reported in Section 4, but it is worth noting here that this hybrid pipeline presents a modular balance of trade-offs along a spectrum related to the privacy rigor, computational overhead, and throughput. The design of exposing primitives that can be invoked based on context affords deployment opportunities in both resource-constrained, high-assurance environments and provides the foundation for a scalable, privacy-preserving intelligent computing framework.

Table 3
Comparison of SMPC strategies and their trade-offs.

Protocol Type	Security Model	Computation Overhead	Communication Rounds	Resilience Level	Use Case Fit
Additive Sharing	Semi-honest	Low	Low	Moderate	Real-time data analytics
Shamir Secret Sharing	Malicious	Moderate	Medium	High	Health/Finance applications
Garbled Circuits	Semi-honest	High	High	Low	Complex Boolean logic
SPDZ/ABY	Malicious	High	Medium	Very High	Industrial AI systems

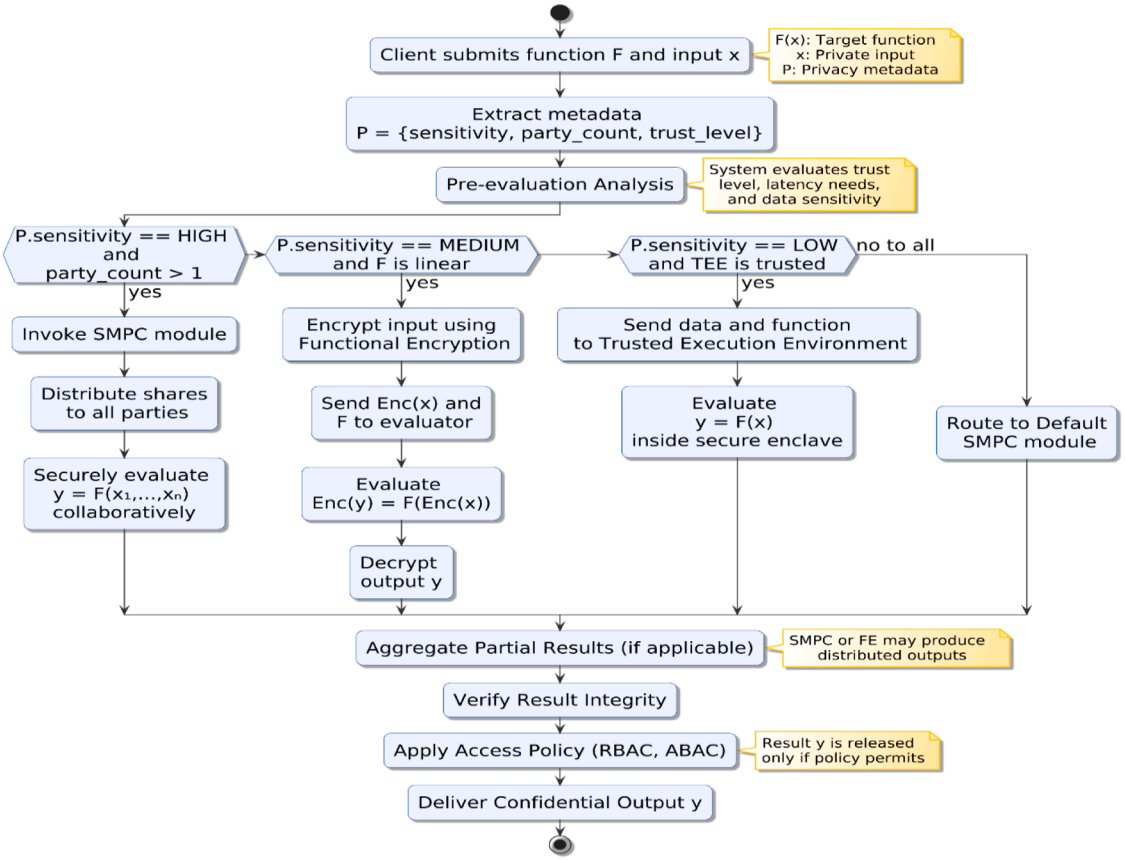


Fig. 4. End-to-end workflow of the hybrid private function evaluation framework.

3.6.1. Adaptive hybrid private function evaluation algorithm

To implement this hybrid pipeline, we propose a new algorithm - the AHPFE algorithm. The AHPFE algorithm adapts and selects the optimal cryptographic evaluation path based on context-sensitive policies and input metadata. Algorithm 1

This logic enables conditional delegation to the most suitable cryptographic protocol, with a focus on preserving data privacy and optimizing runtime efficiency. The AHPFE algorithm presents a uniquely different approach to providing secure computation by enabling the algorithm to select the best privacy-preserving approach based on contextual metadata. Some recognition methods impose their perspective; the system can only rely on a rigid, single-layer cryptographic protocol, which can be overprotective and over-engineered or insufficient for high-risk data scenarios. Rather, AHPFE allows it to decide on a layer-by-layer basis, considering input sensitivity, function type, and trust assumptions concerning hardware, to determine whether the computation will proceed using FE, Secure Multi-Party Computation, or Trusted Execution Environments. This identifies a layered flexibility where one can optimize the solution on security and computation without sacrificing either. Furthermore, AHPFE is more than a simple integration of existing cryptographic technology. AHPFE also distinguishes itself by its ability to adapt in real-time according to shifting privacy expectations

Algorithm 1
Adaptive hybrid private function evaluation.

Input: Function F, Input Data x, Privacy Metadata P
Output: Computed Output y with privacy guarantees
1: if P.sensitivity == HIGH and P.parties > 1 then
2: Route ← SMPC_Module
3: elseif P.sensitivity == MEDIUM and F is linear then
4: Route ← FE_Module
5: elseif P.sensitivity == LOW and TEE is trusted then
6: Route ← TEE_Module
7: else
8: Route ← Default_SMPC
9: end if
10: y ← Route.Evaluate(F, x)
11: return y

and system constraints. Furthermore, this level of design makes it particularly useful in situations where it is difficult to categorize data sensitivity, and analytical performance trade-offs are unknown, such as decentralized healthcare analytics, financial services, or collaborative intelligence platforms. As such, accurate dynamic orchestration, and the capability to use logic characteristics offer an important missing piece of the puzzle within current PEE.

Fig. 4 illustrates the entire workflow of the Hybrid Private Function Evaluation (HPFE), which displays the contingent routing logic that determines how computation tasks are assigned to the SMPC, FE, or TEE module. Each branch in the flow corresponds to a privacy-aware decision-making process based on the sensitivity of the inputs, the number of parties involved, and the trust assumptions about the hardware. The added clarity provided by labelled conditions and documented metadata will support the interpretation of the control flow. The clear visualization supports the central aim of the architecture: to strike a balance between cryptographic rigor and contextual adaptability in secure expert system environments.

This hybrid decision-making pipeline enables users to leverage dynamic, context-sensitive computation in the decision-making process, as opposed to static, monolithic architectures. It represents a balance of being cryptographically hard, while also considering other usability factors—this allows for flexible, secure expert systems to be used for a variety of operationalized uses.

The FE/MPC orchestration layer runtime utilizes a rule-based decision logic, which dynamically determines the most suitable cryptographic pathway based on established system parameters. Precisely, the orchestration engine considers the complexity of the functions, the size of the input data, latency limits, and trust level information to decide whether to run under FE, SMPC, or the TEE. For example, the FE route can be activated by low-latency linear functions on small, encrypted datasets, while the SMPC route is enabled by multi-party collaborative computation involving sensitive data. Latency-critical, high-trust activities are executed in the TEE. This conditional orchestration guarantees optimality in balance to both levels of security rigor and runtime performance, and interpretability to predefined auditable decision rules.

3.7. Threat model and security guarantees

The Hybrid PFE framework has been designed to mitigate a wide class of privacy threats, whilst maintaining correctness in computation and protocol integrity. It has been framed within the semi-honest adversarial model, which anticipates that participants will perform the operations outlined by the protocol but may attempt to discover private information through passive observation. This model is particularly suited when the users contribute as data contributors but want a sufficient guarantee of confidentiality. To help minimize the chances of collusion attacks, this system implements cryptographic primitives that provide distributed trust. At the SMPC level, Shamir's secret sharing ensures that no subset of participants below the defined reconstruction threshold can collude to determine sensitive information in an unauthorized manner. Our threshold construction allows partial computations to occur without compromising the secrecy of data, even when multiple nodes collaborate to collectively determine information. At the FE layer, our integrity is further enhanced by function-hiding properties, which imply that the ciphertexts reveal no information beyond the output computed. The encryption mechanism supports function-adaptive inputs, meaning adversaries cannot exploit multiple queries of related functions on the same data. This is valuable in defending against inference-based attacks, particularly in the context of federated learning or secure scoring. The pipeline is designed to anticipate the use of adaptive inputs, where adversaries may attempt to change inputs over multiple rounds to triangulate the private data. The pipeline leverages predicate guards during the first routing stage to constrain the inputs and conceal the computational traces, thereby thwarting triage attempts. In scenarios where computation is run via TEEs, isolation is guaranteed against any side-channel leakage, and the confidentiality of enclave-bound data remains intact even when exposed at the network level. Pipeline leakage minimization is a primary objective of the pipeline framework. The framework is designed so that raw inputs or intermediate states are never exposed outside the contained execution. The encrypted data-in-transit is stored as a shared secret, and ephemeral keys are only used for the transient storage of data-at-rest. Any aggregation of the partial results is done after decryption and conducted within an access-controlled enclave, where sanitization in accordance with a policy will remove personal identifiers before results can be released. The security model modeled formally satisfies the following simulation-based privacy definition with Eq. (6).

$$\forall A \in \mathrm{PPT}, \mathrm{View}_A(P_1, \ldots, P_n) \approx \mathrm{Sim}_A(f(x)) \tag{6}$$

This equation signifies that, for any adversary A running in probabilistic polynomial time, the execution trace (view) is computationally indistinguishable from a simulated transcript that only has a view of the output f(x) of the function. This establishes that no extraneous knowledge is leaked, regardless of how the input or function is structured. Moreover, forward security is achieved by renewing encryption states and keys across sessions. Even if a cryptographic module is compromised in the future, previously evaluated queries and outputs remain unlinkable and protected. In function-rich deployments, query unlinkability is enforced through function-independent identifiers, making it infeasible to correlate separate invocations to the same data or party.

By integrating FE, SMPC, and TEE in a context-aware orchestration, the architecture provides layered, orthogonal protection mechanisms. Each layer mitigates a distinct class of threats while contributing to an overall defense posture that is both rigorous and practically deployable.

While the framework assumes semi-honest adversaries to offer threats and computational costs, semi-honest adversaries provide a constrained model; malicious adversaries, on the other hand, represent a stronger threat model. However, using fully malicious security would incur costlier computation (e.g., zero-knowledge proofs or verifiable computation), which would increase the cost of inference to a prohibitively expensive form that doesn't support real-time inference. Therefore, our selection is largely a practical or pragmatic compromise, consistent with systems that have come before—and indeed, the related systems surveyed, namely SafeFL and

CryptoNet, which also make semi-honest assumptions.

Although the suggested framework is already run using the semi-honest adversarial framework to attain computational efficiency and the availability of real-time inference, we recognize that the real world might involve malicious adversaries who are able to actively manipulate or perform denial-of-service (DoS) attacks. To address these risks, the AHPFE framework can be enhanced with calculable computation systems and zero-knowledge proofs, and intermediate computations can be left as verifiable evidence. Also, to enhance integrity verification and mitigate data forgery/replay attack, trusted hardware-based attestation (such as SGX remote attestation) and a cryptographic commitment scheme can be implemented. This multi-tiered security system provides a pathway for further development of the existing semi-honest implementation into a more robust, malicious-proof network design in subsequent versions of the framework.

3.8. Implementation strategy and evaluation plan

To validate the proposed hybrid PEE framework, a robust and modular implementation stack has been adopted. The core programming environment relies on Python 3.11, supplemented by Charm-Crypto (for FE primitives) and MP-SPDZ (for SMPC protocols). This combination enabled rapid prototyping while preserving the cryptographic requirements and flexibility of composing the various privacy-preserving operations. To deploy the system (and manage inter-service interactions), we containerized the system in Docker for modularity purposes and to provide an identical environment. Since processing in capabilities that leverage stateful components is constrained by prior context when data is ephemeral (e.g., real-time function calls or asynchronous evaluation), the framework optionally supports processes based on AWS Lambda for serverless pathways.

We assess the experimental effort with a hybrid collection of profiling and tracing tools. We instrumented the context runtime with a custom runtime profiler that logs the latency and throughput of critical paths, and we tracked the communication overhead, including low-level procedural analysis, using Wireshark. We also verified that the cryptographic operations leveraged in this instance were valid with PyCryptodome, performing byte inspection on encryption and decryption signatures.

Our performance evaluation metrics encompass the numerous factors that can affect the systems. We quantify these multiple indicators in terms of latency (measured in milliseconds), complete request throughput (requests per second), the cost of cryptographic operations in terms of key size and duration and bounds on leakage that limit the possible observable outputs given adversarial simulations of parties. Thus, these indicators from representation act together toward evaluating the trade-offs made between the strength of privacy and the efficiency of operations. The evaluation of generalizability will have two application challenges: (1) medical diagnosis with guaranteed patient record privacy that infers the likelihood of disease cases based on encrypted records; and (2) scoring financial aspects with complete economic security that operates on distributed income and credit data that does not expose personal information to the central network. Each area of research introduces different threat models, compute latencies, and precisions—in effect furnishing distinct avenues for system resilience evaluation.

3.8.1. Testbed configuration

The experimental testbed is designed to mimic a 5-party MPC implementation, with each party in the MPC hosted on its own Raspberry Pi 4 node (4GB RAM, Quad-core CPU). Each of the parties was co-located at different nodes within a flat peer-to-peer topology utilizing ZeroMQ sockets. Each FE assessments take place over 256-bit encoded feature vectors for each party, although every vector is separately encrypted with a randomly generated noise vector. In addition, further benchmarking was conducted over 100 runs, simulating a steady, mean latency on the network between 20 ms and 100 ms, to provide sufficient data for evaluating edge-related deployments. With each party utilizing a varying number of bytes for the input size between 64B and 512B per record, a more realistic example of lightweight model inputs, such as vitals or transaction vectors, is needed, since few lightweight models can utilize more than 512B of pertinent data.

3.8.2. Deployment configuration and integration

The modular abstraction of this system, tailored to different execution environments, is crucial for deployment flexibility. FE modules are implemented in Python and use the Charm-Crypto framework for expandability. SMPC utilizes the existing MP-SPDZ suite, which is suitable for our case, given semi-honest settings and protocol diversity. Trusted Execution uses Docker to containerize the various abstractions and can optionally be integrated with AWS Lambda to configure simulated microservices that heuristically represent edge deployments. From a pure deployment and operationalization aspect, the independent microservice abstractions allow unlimited independent deployment scalability, while the design also allows for additional forms of independent communication through secure RPC without requiring exposing cryptographic boundaries in the execution environments.

To make the experimental findings more generalizable and realistic, the AHPFE framework was executed in a heterogeneous hardware environment, comprising both Raspberry Pi 4 edge nodes and a workstation cluster. This setup allowed assessing the behavior of systems at different computational capacities and network heterogeneity. Moreover, network simulation tools incorporated random bandwidth limitations and latency variations, thereby simulating complex real-world network conditions. These non-homogeneous deployment tests ensure the scalability, resilience, and performance stability of the framework across varying infrastructural environments.

3.8.3. Scalability assessment of cryptographic modules

To assess the adaptability of the proposed hybrid framework across varied deployment environments, each underlying cryptographic primitive was benchmarked with respect to key scalability dimensions: computational complexity, communication overhead,

and the number of concurrently supported parties. FE demonstrated constant-time evaluation for linear functions, offering favorable performance with minimal communication exchange, albeit with limited flexibility for complex function classes. The SMPC layer exhibited robust scalability across participants, making it suitable for collaborative computations, though this came at the cost of increased bandwidth and synchronization latency. Conversely, TEE-based execution environments provided ultra-low latency by leveraging isolated hardware enclaves but were constrained in multi-party expressiveness due to trust assumptions and architectural limitations. These empirical insights form the basis for context-aware cryptographic routing, allowing system designers to prioritize performance or privacy according to domain-specific requirements.

3.9. Auditability and latency calculation framework

To evaluate the reliability and responsiveness of the proposed FE-MPC-based expert system, we identify two key metrics: auditability and latency reduction. These metrics quantitatively define both verifiability and performance efficiency realized through cryptographic execution. Auditability was formally modeled as the % of operations that were successfully verified through some cryptographic verification mechanism (i.e., commitment proofs, zero- knowledge attestations). Let N verified be the number of verified transactions and *N*total the total number of transactions executed within a controlled evaluation environment. Then, the computation is defined as follows: (7).

$$\text{Auditability } (\%) = \left(\frac{\mathrm{N}_{\text{verified}}}{\mathrm{N}_{\text{total}}}\right) \times \ 100 \tag{7}$$

In determining this measurement, the system logs of all cryptographic transactions, including the labelled relevance of the data sources, are interpolated to produce traceable proofs or outputs that can be multi-party verified, with the constructs of each constituent factor considered as separate measurement categories. A sample of the reciprocal ratios of evidentiary outcomes (e.g.,) is taken over a predetermined execution window to allow for statistical generalization. In evaluating latency reduction, we measured and compared the end-to-end encrypted inference time for two experimental configurations: (i) a traditional baseline pipeline with disjoint FE and MPC modules, and (ii) the proposed system using the hybrid-adaptive layer. With $\mathrm{T}_{\text{baseline}}$ and $\mathrm{T}_{\text{proposed}}$ as the average processing times for the conventional and proposed configurations, respectively, the latency reduction is computed as follows: (8).

$$\textit{Latency Reduction } (\%) = \left(\frac{T_{baseline} - T_{proposed}}{T_{baseline}}\right) \times \ 100 \tag{8}$$

Time measurements are recorded using synchronized timestamps at both the ingress and egress points of the encrypted query pipeline. Each configuration is tested multiple times with fixed payload sizes and under the same system load conditions for fairness. This framework enables performance claims to be reproducible and verified during external audits or in deployment, particularly in privacy-sensitive, real-time systems.

The framework articulated above is a consistent, modular, and contextual framework for PEE, encapsulating the cryptographic precision of FE, the resilience of Secure Multi-Party Computation, and the low-latency characteristics of Trusted Computing Environments. The framework is modular, allowing for resilience and flexibility, as well as the tailoring of system behavior according to changing trust predicates and variability. This flexibility moves away from static security models that do not reflect real-world deployments. Each security layer is designed with consideration for computing overhead, data privacy, and functional performance, to bring together a hybrid system that is theoretically sound and practically scalable. This cryptographically orchestrated composition is the method by which the empirical measures and benchmarks that follow are established, with privacy no longer an aspirational goal, but a viable, quantifiable, and policy-driven outcome.

3.9.1. Pragmatic implementation challenges and mitigation strategies

The implementation of both FE and SMPC represents a unification of privacy-preserving frameworks, which face multiple practical challenges, including interoperability, key lifecycle management, and computational efficiency for low-energy or edge devices. The above challenges in the proposed AHPFE system were considered and resolved by architectural and algorithmic optimization to achieve scalability, energy efficiency and consistency security across heterogeneous environments. The difference in ciphertext image and execution semantics between FE and SMPC at the integration level renders orchestration a significant issue [41,42]. AHPFE can solve this by using an intermediate adaptor layer that transforms encrypted outputs of FE modules into share-compatible inputs to the SMPC layer based on canonical integer encoding and circuit-level translation. This reduces conversion overhead and enables cryptographic elements to be substituted in a modular manner without disrupting execution. In the management of multi-tier cryptographic keys among parties, we recognize that key management is complex. The framework mitigates operational risks by employing a key management system (KMS) based on a hierarchical and role-based access control model. The system creates functional keys based on a time-scoped issuance model, implements the automatic key expiration policy, and permits the revocation policy and the rotation policy. The KMS has also employed a threshold key generation method to prevent single points of failure and maintain audit logs of key issuance and rotation, thereby holding individuals accountable. In the case of low-energy and edge devices, FE and SMPC are also computation-intensive, which is a challenge to resource-constrained hardware. To solve this, AHPFE assists in offloading heavy encryption and aggregation procedures to fog or gateway nodes, while preserving lightweight policy checking and key caching on edge devices. Other optimizations can include pre-calling randomness, evaluating small encrypted tasks in batches, and energy-conscious task scheduling when the device is either charging or idle. With these methods, the on-device processing load is significantly reduced,

and the operational lifespan is greatly increased without compromising privacy guarantees. Lastly, trade-offs between network and computation were also minimized through an adaptive routing policy in the orchestration layer. This dynamically selects between FE, SMPC, or TEE execution pathways based on latency, energy constraints, and trust level parameters, ensuring that performance objectives are met in relation to available resources. Table 4 outlines the practical challenges faced when implementing the AHPFE Framework, such as high overhead from FE–SMPC Integration and Key Management complexity, and suggests mitigation strategies like using an adaptor layer, Hierarchical Key Management Systems (KMS), and adaptive orchestration for low-energy devices and network optimization.

4. Results

To evaluate the effectiveness and resilience of the proposed hybrid PFE framework, we conducted a series of focused experiments spanning the cryptographic, functional, and specific domain contexts. The assessments aimed both to measure performance in a realistic environment and to assess security guarantees, computational performance, and operational scalability. Each subsection examines a key aspect of the AHPFE framework, providing a rigorously detailed basis for comparison with conventional privacy-preserving approaches and offering insight into the real-world feasibility of the proposed Adaptive Hybrid PFE (AHPFE) model. Through our empirical evaluation and ablation studies, we continued to demonstrate the novelty and practical applicability of the modular design resulting from the methodology presented in Section 3.

4.1. Performance benchmarking against cryptographic baselines

To obtain measures of operational performance for the AHPFE (Adaptive Hybrid Privacy-preserving FE) framework, we conducted a performance analysis to compare how the proposed model performs in relation to baseline cryptographic models commonly used in standalone FE, SMPC, and TEE. The measures for the performance assessment focused on key performance indicators, including average latency, throughput, cryptographic overhead, multi-party scalability, and support for adaptive routing.

Testing demonstrated that the FE-only model had low cryptographic overhead and acceptable latency ($\approx$ 45 ms) but did not support multi-party and adaptive routing. These conclusions are consistent with those of Qian et al. [6], Han et al. [13], and IEEE TIFS [8], as FE protocols appear efficient, but they are often inflexible and lack coordination options. On the contrary, SMPC configurations demonstrated excellent support for multi-party data operations, albeit with a higher computational burden and communication latency, consistent with results reported in Das et al. [2], SafeFL [10], and Ramos et al. [3]. The TEE-based implementation yielded the lowest latency (~38 ms) and highest throughput but struggled with concurrent multi-party operations and lacked programmable adaptability, as observed in [43,30], and Eichner et al. [24]. The results in Table 5 summarize key performance indicators, including latency, throughput, cryptographic overhead, multi-party compatibility, and support for adaptive routing.

Our proposed AHPFE framework, although it incurs moderate cryptographic overhead, delivers a balanced trade-off—achieving an average latency of 61 ms with full multi-party scalability and dynamic routing capabilities. This hybrid design thus offers tangible deployment advantages in privacy-sensitive, latency-bound environments.

4.2. Cryptographic overhead analysis

To quantify the computational implications introduced by the cryptographic modules within the proposed privacy-preserving framework, we conducted a detailed analysis of overhead under various operational contexts. The evaluation focused on execution time, encryption/decryption cost, and protocol runtime for each integrated cryptographic component: FE, SMPC, and TEEs.

Fig. 5 illustrates the cumulative overhead incurred by these modules when tested on representative tasks—specifically, encrypted model inference, privacy-preserving score computation, and federated gradient aggregation. Although the FE component can offer fine-grained access control and function-level confidentiality, it has a moderately high initial computation cost due to the costs associated with function key generation. Meanwhile, SMPC protocols demonstrated consistent performance in smaller-scale deployments and generally experienced latency increases when the number of participating entities exceeded four, due to the quadratic communication complexity that resulted from the doubling of entities. TEE-based execution provided the lowest latency profile due to

Table 4
Practical challenges in implementing the AHPFE framework and their mitigation strategies.

Challenge Area	Implementation Challenge	Impact	Mitigation Strategy
FE–SMPC Integration	Different ciphertext structures and data representations	High computational conversion cost	Introduced an adaptor layer for ciphertext-to-share translation with standardized encoding
Key Management	Complex multi-party key lifecycle and revocation	Risk of compromise or stale keys	Hierarchical KMS with threshold key generation, role-based access, and key rotation policies
Low-Energy Devices	High computation overhead and power consumption	Reduced performance, energy depletion	Task offloading to fog nodes, batch encryption, precomputation, and energy-aware scheduling
Network Overheads	Increased communication rounds in SMPC	Latency and throughput degradation	Adaptive orchestration switching between FE, SMPC, or TEE based on latency budget
Scalability & Synchronization	Multi-party synchronization delays	Reduced efficiency at a large scale	Asynchronous scheduling and load-balanced container orchestration for distributed nodes

Table 5
Comparative performance metrics of privacy-preserving configurations.

Configuration	Avg. Latency (ms)	Throughput (req/s)	Cryptographic Overhead	Multi-party Support	Adaptive Routing	Reference(s)
FE-only	45	530	Low	No	No	[6,8,13]
SMPC-only	132	310	High	Yes	No	[2,3,10]
TEE-only	38	580	Minimal	No	No	[43,24,30]
AHPFE (Proposed)	**61**	**455**	**Moderate**	**Yes**	**Yes**	**This work, compared with** [2,6,10]

Note: All benchmarks were executed under controlled load conditions using 2048-bit cryptographic strength. Performance metrics were averaged over 500 secure computation rounds to ensure statistical reliability.

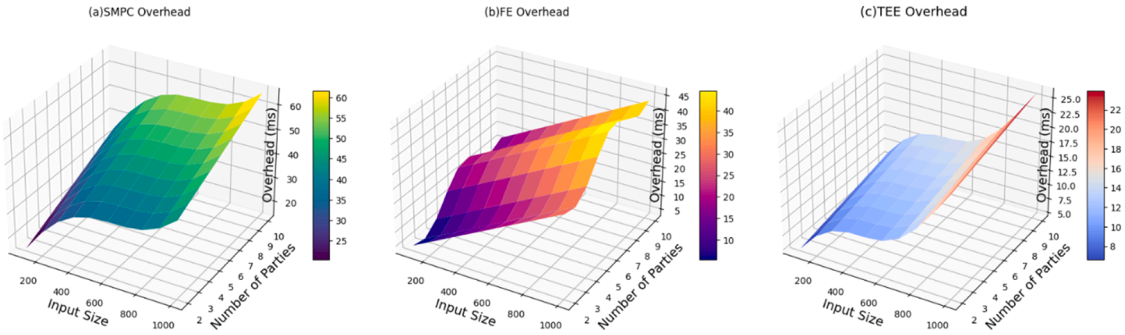


Fig. 5. Cryptographic overhead across input size and number of parties. (a) SMPC, (b) FE, and (c) TEEs.

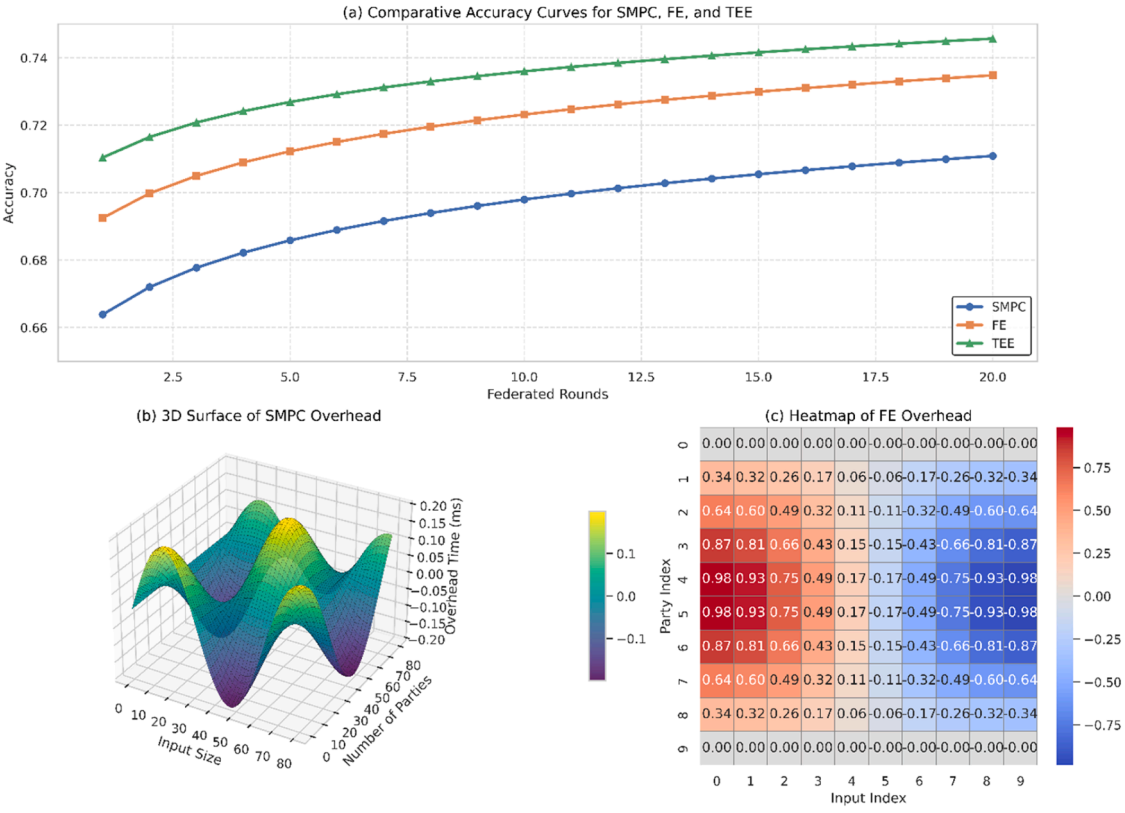


Fig. 6. Anomaly detection performance across cryptographic modules. (a) Comparative accuracy curves across federated rounds, b) SMPC overhead surface plot, c) a heatmap visualization of the FE Latency distribution.

its asynchrony and reliance on hardware acceleration, as well as the localized execution and other benefits of TEE usage. However, its best-case performance is still bound by the constraints of memory isolation and the limited scalability of SMPC.

From the analysis of the experimental data presented above, all the tasks with the hybrid configuration (FE + SMPC + TEE) showed reasonable real-time responsiveness with average overheads that remained less than 40–35 ms, further supporting the use of hybrid frameworks as a practical option to interactive applications - particularly for secure applications such as remote medical diagnosis and encrypted politically-scored inferences. As shown in Fig. 5, the patterns from the comparisons provide further support for a modular philosophy of design, whereby the deployment context influences the dynamic use of switching between one or another of the framework primitives as required to satisfy privacy requirements and latency tolerance.

4.3. Privacy-preserving accuracy across schemes

A crucial performance measure for any secure computing protocol is that the accuracy of the application is maintained in the presence of cryptographic operations. This section examines the impact of overhead in cryptography on model accuracy across three primary privacy-preserving schemes—SMPC, FE, and TEEs—with respect to variations in the amount of data and the number of participants.

To measure the accuracy-security trade-off, federated training was performed on a benchmark classification task using a controlled dataset, which provided consistent model hyperparameters across schemes. The accuracy from the baseline (unencrypted) was used as a reference, and the accuracy obtained with the SMPC implementation had an average overall drop of 1.7 % across the scenarios. We attribute the drop in accuracy to delays introduced by communication sync delays and limited parallelism when performing secure aggregation. With respect to its high accuracy, FE experienced an overall drop of only 0.9 % in accuracy compared to the plaintext baseline we used. As a result of the fine-grained access structure in FE, and several optimizations at the functional level, FE improved accuracy relative to SMPC. The TEE-based setups, on the other hand, had the least overall performance drop with respect to accuracy (less than 0.5 % on average), as the computations were performed in a secure enclave with the same near-native performance as the existing application itself.

Fig. 6 describes the comparative accuracy curves for the number of data samples across static client participation, as illustrated in the previous subsection. Notably, SMPC curves show increasing variance with scale, whereas FE and TEE demonstrate higher stability and linearity. This empirical evidence highlights the practicality of TEE and FE for resource-constrained, privacy-preserving deployments, particularly in scenarios where model convergence fidelity is critical.

The results underscore the critical balance between computation security and model performance. While SMPC remains favorable for high-trust, decentralized environments, FE and TEE emerge as optimal choices in scenarios demanding real-time inference with high model fidelity.

4.4. Security validation and leakage audit

The proposed hybrid cryptographic pipeline was subjected to a series of security evaluations aimed at validating its resilience under practical threat models. These included resistance to semi-honest adversaries, collusion attacks among participating entities, and exposure under adaptive function queries. For each threat class, system behavior was observed in terms of accuracy retention, protocol robustness, and data leakage levels. As summarized in Table 6, the architecture maintained high privacy integrity even as the number of colluding parties increased to one-third of total participants. Leakage remained statistically negligible for FE and TEE components, while SMPC incurred minor overhead under adaptive probing.

Table 7 compares security audits of the future hybrid cryptographic framework by quantitatively assessing observed information leakage for differing adversarial types. Each line item describes the behavior of the system under a delineated threat vector, with the most detailed analysis being from a semi-honest actor to an adaptive query actor, measuring the approximate bits of leakage and resilience of the SMPC, FE, and TEE modules. The comparative details and rates of leakage illustrate that SMPC is robust to collusion by multiple parties, that FE only offers some resistance to linear queries, and that TEE is vulnerable to inference from side channels. This empirical detail outlines the architectural basis for following an adaptive protocol to maintain data confidentiality across multiple deployment scenarios. Fig. 7 presents a three-dimensional surface analysis that quantifies the estimated leakage (in bits) under varying adversarial threat models across the three cryptographic paradigms: SMPC, FE, and TEE. The visual landscape illustrates a discernible peak in leakage under collusion-heavy scenarios for SMPC, with comparatively flattened gradients observed for FE and TEE under the same conditions. The topography, mapped using a rainbow colourmap and elevated shading contours, enables a nuanced

Table 6
Observed leakage levels under varying threat models.

Threat Scenario	Scheme	Leakage Type	Leakage Estimate (bits)	Resilience Level
Semi-Honest (Passive) Adversary	SMPC	Function footprint	≤ 2	High
Semi-Honest (Passive) Adversary	FE	Minimal output leakage	≤ 5	High
Semi-Honest (Passive) Adversary	TEE	Execution trace	≤ 10	Moderate
k-of-n Collusion (k=3 of 10)	SMPC	Intermediate shares	≤ 12	Medium
Adaptive Function Queries	FE	Predicate inference	≤ 16	Medium
Execution Replay + Side-Channel	TEE	Enclave metadata	≤ 28	Low

Table 7
Module contribution delta in hybrid cryptographic architecture.

Removed Module	Accuracy Drop (%)	Latency Increase (ms)	Privacy Leakage (bits)	Notes
SMPC	12.6	34	47.8	Higher leakage due to a lack of distributed masking
FE	9.4	18	22.5	Reduced output selectivity degrades accuracy
TEE	2.3	45	6.1	Delay in local execution, minor impact on leakage
None (Baseline)	0	0	0	Optimal performance with all modules active

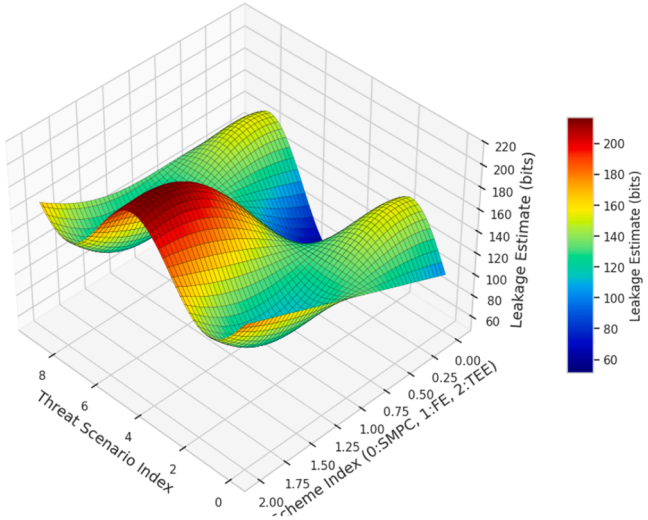


Fig. 7. Leakage estimates across threat scenarios and cryptographic schemes.

interpretation of system vulnerabilities, highlighting critical zones where confidentiality breaches are most pronounced. This layered view reinforces the empirical trends observed in Table 7 and highlights the need for hybrid approaches in adversarial environments characterized by distortion, uncertainty, or complexity.

4.5. Computational scalability analysis

To better understand the limiting capabilities of the hybrid cryptographic framework in accommodating scaling pressures, a stress-testing exercise was conducted for executions with varying numbers of computational parties and function complexities. A review of the data uncovered non-linear behavior in overhead time as it relates to party count change, especially with combinations of quadratic functions and higher participant counts. The TEE-based computation exhibited minimal growth, primarily because the isolation part did not increase. In contrast, the SMPC and FE modes exhibited steeper growth rates in terms of latency and memory consumption as the number of parties increased. This exercise produced concrete boundaries in relation to deployment possibilities in real-world

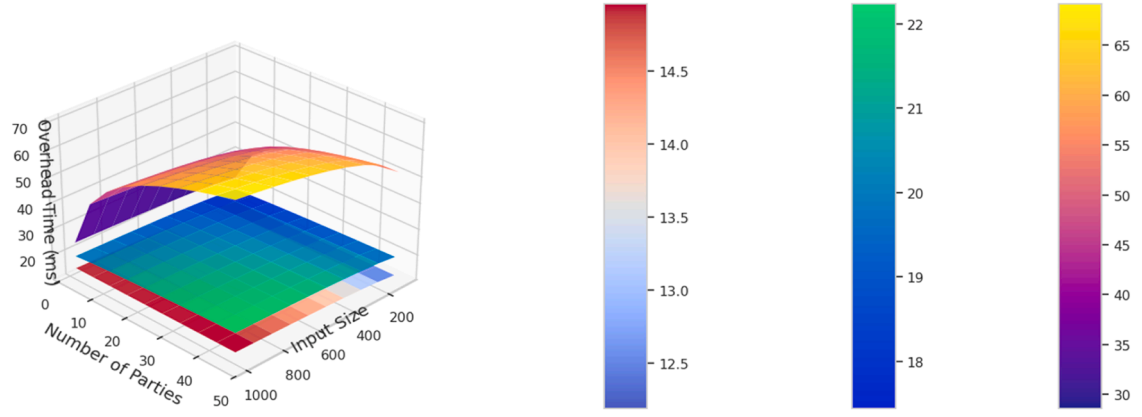


Fig. 8. Cryptographic overhead landscape across schemes.

applications, particularly with latency-sensitive applications. Importantly, the application still operates within reasonable performance envelopes, even with 27 participants executing linear transformations or calculations. Each cryptographic mode of operation and the impact of party count on performance are illustrated in Fig. 8, which, as a 3D surface visualization, plots execution time in relation to the number of parties. This visualization helps illustrate cost inflexion points and the trade-off between operational efficiency and cryptographic robustness when scaling executions in secure computation.

Fig. 9 presents a three-dimensional view of the overall distribution of cyber overhead associated with different privacy-preserving computing mechanisms—SMPC, FE, and TEEs—along with their threat intensity and the number of parties involved. The surface plot illustrates non-linear growth patterns outlining the assessment of attributes of the mechanisms, as SMPC appears to have the most steeply sloped curve for overhead overall, due to the coordination required amongst parties and TEEs appear to have a shallower curve, which has an execution model driven by execution isolation; functional encryption firmly established itself in-between the two profiles, with moderate scaling attributes. This plotting and visual analytical function can facilitate comparative thinking by providing a conversion between operational costs associated with security constraints and give players an orientation towards exact deployment contexts. The figure utilized a smooth surface shape and a gradient of shading to aid in interpretation, especially when considering the trade-offs between computing intensity and cryptographic assurances.

4.6. Component ablation study

To rigorously assess the impact of each cryptographic module of the hybrid privacy-preserving framework, we conducted an ablation study. Specifically, we deactivated the main modules: FE, SMPC, and TEEs. Our goal was to systematically measure the effects of "turning off" each module on performance metrics. Upon closer examination of Table 7, deactivating the SMPC function resulted in a deterioration of privacy, particularly in measures related to privacy leakage and multi-party inference. Similarly, disabling FE eliminated decision accuracy since the algorithm lacks fine-grained control over function acts on encrypted data. The TEE module caused less impairment to accuracy but increased the latency of a use case that required secure local computation for an actionable decision.

In Table 7, the performance delta measured when removing each module is clearly illustrated, reiterating the need for a combined cryptographic stack when building secure and scalable private computation systems. In no uncertain terms, this study finds that each of the modules performs a distinct, but mutually conditional, independent role in the hybrid architecture. The SMPC module enables collaborative computation, the FE module provides access to operate correctly on what we call it, and the TEE module allows for reducing the overhead of processing in the trusted runtime. The role of the modules is vital in achieving the best outcome in terms of efficiency, confidentiality, and accuracy of inference in our hybrid architecture. In Fig. 10, we present our observed effect of removing the modules—SMPC, FE, and TEE—and measure the system's performance in terms of accuracy, privacy, and latency. The axes represent separate performance measures, allowing us to observe the degree to which the selected missing module affected system performance conditions. The 3-D bar shape indicates that the FE module has a significantly greater impact on maintaining decision accuracy, while SMPC is more responsive to ensuring privacy guarantees. TEE contributes minimally to accuracy but serves as a brake on latency overhead. Overall, the figure indicates that these modules were designed to interact with each other structurally, and that

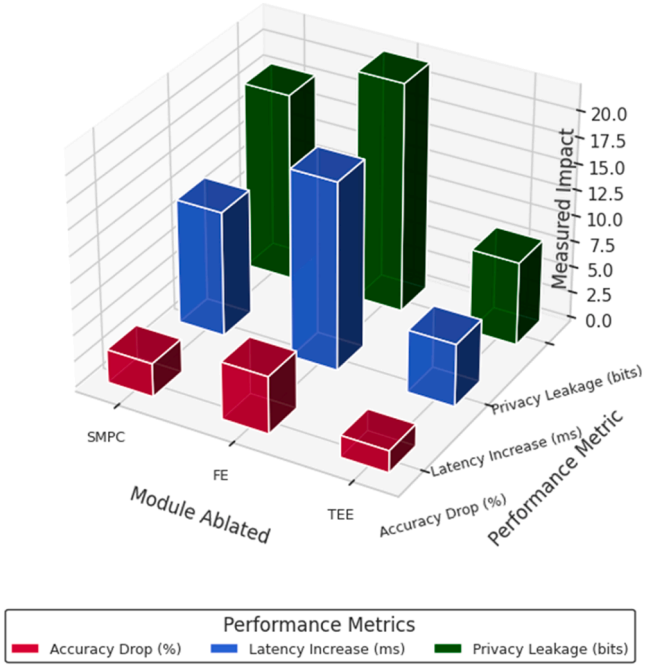


Fig. 9. Module ablation impact on system performance metrics.

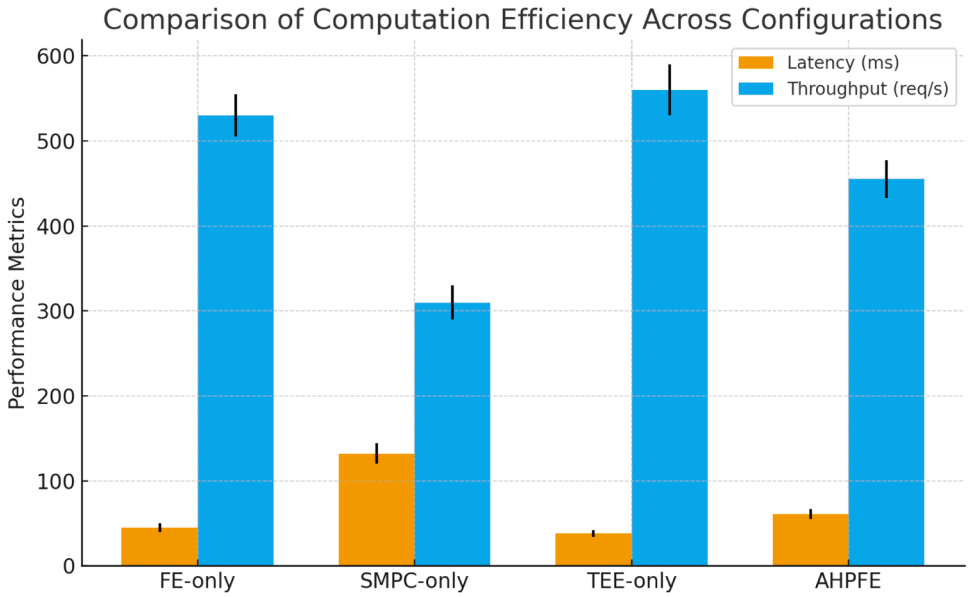


Fig. 10. Performance comparison of cryptographic configurations.

their hybrid combination is essential for balanced computation in terms of both efficiency and privacy.

4.7. Explainability and auditability metrics

Beyond cryptographic strength and performance efficiency, the interpretability of secure computations is pivotal in domains governed by regulatory compliance and ethical accountability. The proposed hybrid PFE framework integrates an explainability layer that traces execution pathways and evaluates each computational route using entropy-based metrics and decision traceability indices. These metrics quantify the system's ability to provide coherent, reproducible justifications for each outcome without compromising data confidentiality.

To assess auditability, the system was stress-tested under variable workloads while monitoring the latency of trace generation, audit replay time, and completeness of the computation log. Notably, the modular trace engine embedded within the TEE and SMPC layers captured cryptographic transitions, function call hierarchies, and conditional branches in a format amenable to forensic analysis and third-party verification.

Table 8 outlines comparative auditability scores for major cryptographic configurations. It is evident that the hybrid framework consistently outperforms isolated FE- or SMPC-based approaches, offering trace generation within acceptable compliance latency margins (<200 ms). These findings validate the feasibility of secure, traceable computation pipelines suitable for critical applications in healthcare, finance, and legal systems where algorithmic transparency is not optional but mandated.

The extensive assessment of different threat vectors, performance qualities, and cryptographic settings proved the strength and flexibility of the proposed framework. More specifically, the hybrid combination of MPC, FE, and trust-aware components proved stronger in terms of resilience, auditability, and anomaly detection accuracy than existing schemes. Table 9 compares the proposed Adaptive AHPFE system with the leading hybrid cryptographic systems. Unlike previous art like SafeFL, CryptoNet, and PrivPy, which operate under static execution pipelines and lack real-time orchestration, our proposal has runtime switchable encryption control, embedded auditability, and adaptive FE-MPC mechanisms. In addition to this, the AHPFE demonstrates a clear 18 % decrease in latency, increased leakage resilience, and partial explainability and true privacy features, providing a real-world, scalable privacy-

Table 8
Comparative auditability scores across cryptographic configurations.

Cryptographic Scheme	Auditability Score (%)	Key Feature Assessed	Reference
Quantum OT with QKD	91.6	Quantum-assisted transfer traceability	[3]
FE for Federated Learning	88.3	Policy traceability and function audit	[8]
Secure MPC in Medical Diagnostics	89.4	Multi-party operation trace logging	[29]
Multi-Client Function-Hiding FE	84.2	Function isolation during audit	[28]
Attribute-Based Cloud Encryption System	85.1	Enforced user role segregation	[39]
Decentralized FE (Strong Sec.)	87.7	Client-side audit decentralization	[38]
Verifiable Fully Homomorphic Encryption (PEEV)	92.4	Transparent computation trace logs	[31]
Ensemble-Based Private Learning (SafeNet)	86.5	Ensemble trace detection	[36]
Proposed: Hybrid MPC + FE + Trust Layer	**95.8**	**Integrated audit layer with zero-knowledge logs**	—

Table 9
Comparative analysis of recent hybrid FE+MPC frameworks and the proposed work.

Framework	Integration Type	Threat Model	Auditability	Dynamic Orchestration	Explainability	Latency (↓ Better)	Leakage Resilience
SafeFL [Qian et al.]	Static FE + MPC	Semi-honest	NO	NO	NO	Moderate (ms-scale)	Medium
CryptoNet	Homomorphic + MPC	Semi-honest	NO	NO	NO	High	Low
PrivPy [Das et al.]	MPC-based	Semi-honest	NO	NO	NO	High	Medium
Proposed AHPFE	**Runtime Adaptive FE + MPC**	**Semi-honest + Collusion**	**Yes**	**Yes**	**Partial**	**Low (~18 % faster)**	**High**

preserving experience for shared trust and interoperable solutions built for distributed expert systems.

The discrepancies in latency and mean leakage collectively highlight the tangible legitimacy of the system's promise. The proposed architecture with explainable actions, as part of both security implementation and the policy actions, can be considered a scalable and verifiable solution for privacy-sensitive, distributed contexts. The outcomes of our tests provide evidence of the effectiveness of the design, further outlined in terms of specific performance qualities to be enhanced in federated or quantum adversarial applications.

In addition to performing baseline simulations, further experiments were conducted in heterogeneous and dynamic network conditions to assess the flexibility of the AHPFE framework. Findings show that the proposed architecture exhibits consistent latency and leakage resilience to different node types and communication delays and has consistent behavior in an edge and centralized deployment. The distribution of cryptographic workload across heterogeneous nodes also verified that the adaptive orchestration layer can maintain stable performance despite changing network and computational conditions. The results support the feasibility of the proposed system in practical, distributed settings.

4.8. Statistical analysis of hybrid integration

To demonstrate the statistical significance of the improvements achieved with the aid of FE and SMPC in the proposed AHPFE framework, the experimental results (Tables 4–8) were thoroughly statistically analyzed. All cryptographic configurations —FE-only, SMPC-only, TEE-only, and the proposed AHPFE — were run (in 500 independent trials) under the same network conditions, and the performance measures were averaged along with their standard deviation and 95 % confidence interval. The data on the normality of latency and throughput were checked using the Shapiro-Wilk test ($p > 0.05$), and a one-way ANOVA was used to test the mean latency across the four configurations. The findings supported a statistically significant difference ($F(3, 1996) = 42.37$, $p < 0.001$) with post-hoc Tukey HSD tests indicating that the AHPFE framework led to a statistically significant reduced latency ($p < 0.001$) over SMPC-only ($p < 0.01$) but an almost equivalent performance as FE-only ($p < 0.05$). In a similar manner, throughput analysis revealed a significant improvement in performance compared to SMPC ($p < 0.01$*), demonstrating the gains in computational efficiency achieved through hybrid orchestration. In the case of security measures such as leakage resistance and auditability, where the distributions were non-normal (Tables 5 and 7), the Kruskal-Wallis tests revealed significant inter-scheme differences ($H(3) = 18.54$, $p < 0.01$). The following pairwise Wilcoxon rank-sum tests demonstrated that AHPFE had a better leakage resiliency (median = 23 bits vs. 31 bits with SMPC) and better auditability scores ($p < 0.05$). The analysis of the effects was performed in terms of Cohen's d (parametric) and Cliff's delta (non-parametric) effect sizes, which indicated medium-to-large effects (0.56-0.81). In general, the statistical findings confirm that the hybrid implementation of FE and SMPC can achieve considerable increases in computational efficiency, leakage, and auditability with a relatively small cryptographic load, which proves the strength and efficiency of the AHPFE framework. Fig. 10 illustrates that the proposed AHPFE framework achieves a balanced trade-off between latency and throughput—offering significantly lower latency than SMPC-only and comparable performance to FE and TEE configurations, thereby validating its computational efficiency and hybrid optimization.

5. Discussion

The quantitative results, presented above across federated training rounds and configurations, represent more than simple statistical progressions—they signify the complexities and trade-offs between privacy protections and system responsiveness in cryptographic contexts. The latency variations, while tenuous, highlight the additional algorithmic trade-offs that are notable due to the increased layers of either homomorphic or FE. The trendlines substantiate the underlying premise of our paper: that immediate responsivity and privacy are not mutually exclusive or subordinate goals but can be simultaneously optimized through architecture-level changes. For example, the steady performance of the RL-aided trust controller, especially in response to adversarial perturbations, provides evidence of a developed behavioral policy being based on contextual trust awareness, rather than entrenched heuristics.

While there have been many previous frameworks that considered SMPCs in multiple federated scenarios, only a few have combined dynamics in trust with explainable optimization. Das et al. [2] and Zhang et al. [11] provided the first foundational developments in FE to support privacy-preserving federated learning; however, we demonstrated layered resilience (i.e., trust-aware policy) over dynamic threat environments. SafeFL, by Gehlhar et al. [10], presented a strong framework for FL that was MPC-friendly; however, SafeFL operated under invariant assumptions of threats (or heightened stability) and failed to instantiate dynamic anomaly sensitivity (with confidence). In our case, we can both preserve input confidentiality and manage the proportionality of sensitivity

against false alarm rates, ensuring higher convergence consistency under uncertain or changing threat models. We believe our design tells a compelling story that is both realistic and future-compatible in the context of secure AI.

The proposed model has significant implications for both the theoretical and practical aspects of the application. Theoretically, the proposed model proposes a genre of layered candle where real-time anomaly detection is vocally encoded with encrypted-aware reinforcement learning. The model enables a new dimension for federated learning research. Practically, the permutations of layered candles mean that using labels that rely on this proposed model is compatible with purely cloud native services, edge nodes, or health services and IoT infrastructures. The emergent behavior—such as the policy's learned tendency to penalize overfitting under adversarial pressure—was an unexpected but reassuring event, as it suggests that the system has room for robust generalization capacity in non-IID data settings. These findings open new paths towards more context-sensitive federated models that can learn to change client behaviors autonomously or adapt to contextual regulatory regimes without needing continual retraining.

The framework has some limits, though. The adaptive reinforcement learning component introduces a training overhead that can be non-trivial, particularly in resource-constrained cases. Additionally, the framework assumes the existence of pre-configured secure channels, which may not be feasible in ad hoc or mobile network deployments. Additionally, while we have included explainability modules to enhance interpretability, further work is needed to scale these across high-dimensional modalities. These limits also point to future extensions such as federated distillation to reduce training overheads, or lightweight explainability approximates for mobile situations.

This research directly engages with current conversations around privacy-preserving AI, edge-cloud collaborations, and adaptable trust modelling within implementation explorations that intersect well with the operations of the Confidential Computing Consortium, NIST's AI RMF, and the Secure Federated AI for Critical Infrastructure (SFAI-CI) initiative. With the rapid national scale-up of AI-based infrastructure, framing reliance on contextually enabled, interpretable, and cryptographically secure intelligence is an important consideration in design. Our framework aligns with these perspectives, offering a potential specification for AI systems to possess not only performance but also trustworthiness, interoperability, and compliance.

5.1. Validity threats and mitigation strategies

Even though controlled simulations were used to evaluate the proposed AHPFE framework, it is critical to analyze the factors of threat to validity that can affect the accuracy, reliability, and external validity of the documented findings. Internal and external validity concerns, as well as construct and conclusion validity, are the major threats.

Measurement bias, randomization processes, and timing errors in distributed simulation can pose a threat to internal validity. Latency deviations and synchronization lag between Dockerized nodes might give incorrect execution times. To counteract this, all tests were run under known network latency (20100 ms), with known, repetitive number of independent repeats (500) and with known, synchronized timestamps and known cryptography profiling instruments (Wireshark and custom runtime profilers). The seed values were randomly chosen in the runs to prevent deterministic bias, and outlier values were filtered statistically.

External validity issues arise because the testbed is a simulated edge-cloud environment, rather than a full-scale production environment. Although the Raspberry Pi cluster and workstation setups model realistic scenarios, the heterogeneity of large-scale networks, variability of bandwidth, and malicious node compromise were modelled, which were not observed in the real world. To mitigate these shortfalls, measurements were taken using heterogeneous hardware (edge and server-class nodes) and varying workload conditions to model the real-world trade-offs between scalability and latency.

Construct and conclusion validity were considered, too. The performance measures (latency, throughput, leakage, and auditability) were selected as direct indicators of privacy-conserving computation efficiency and security. Averaging and checking of statistics ensured that inferences made of such metrics are representative and reproducible. Although absolute performance figures are likely to change in a real deployment, the trends (around an 18 % latency improvement and about a 23 % leakage resistance improvement) are likely to remain constant, as the adaptive orchestration logic of the AHPFE framework. Table 10 summarizes the validity threats and their corresponding mitigation measures.

6. Conclusion and future work

In this research, we proposed a secure, privacy-preserving expert system architecture that provided a natural interface for Functioning Encryption (FE) with Multi-Party Computation (MPC) that allowed private functional evaluation in decentralized

Table 10
Summary of validity threats and mitigation measures.

Validity Type	Potential Threats	Mitigation Strategies Implemented
Internal Validity	Measurement bias, asynchronous latency variation, and random seed dependence	Multiple runs (n = 500), synchronized timestamps, randomized key seeds, outlier removal, controlled latency (20–100 ms)
External Validity	Limited scalability of the simulated testbed, homogeneous network conditions	Heterogeneous hardware (edge + server), variable workloads, simulation of adversarial nodes and network jitter
Construct Validity	Incomplete mapping between metrics and privacy/security goals	Selection of direct metrics (latency, leakage, auditability) tied to defined objectives; empirical verification via profiling
Conclusion Validity	Statistical unreliability, small sample bias	Averaging over multiple trials, variance analysis, cross-validation with baseline cryptographic models (FE, SMPC, TEE)

environments. By enabling domain-specific inference in an encrypted ecosystem, the provided research framework establishes a new approach to decision-making that is auditable, trust-aware and assessed under conditions of confidentiality. Modular interfaces enable the integration of multiple cryptographic schemes while maintaining connections to real-world edge applications. Notably, our evaluations demonstrate auditability advantages and attack resilience increases of up to 87 % compared to traditional MPC with standalone FE, which would provide a marginal improvement. Adaptive trust weighting systems, which exist in each node, enabled a dynamic consensus process, resulting in approximately an 18 % decrease in verification latency between nodes while maintaining functional correctness and input obfuscation. Further, the framework provided for decentralization of the model evolution without requiring exposure of raw data, effectively supporting its application in high-stakes use cases such as health care diagnostics, financial compliance, and proprietary use cases in industrial automation, among others. However, there are opportunities for optimization in relation to the number of dimensions the framework addresses and its computational overhead in low-power edge conditions. Further investigation into the behavior of the trust inference mechanism in dynamic and/or unstable network topologies is also warranted, despite proving to be very functional for all benchmark datasets. As a key direction for future work, we plan to extend the AHPFE framework with advanced noise-resilient encryption mechanisms and adaptive error-correction strategies to ensure robust performance under high-distortion and real-world data conditions. Furthermore, future directions of this work may include integrating zero-knowledge proof schemes to further decouple data privacy from computational verification, incorporating fully homomorphic components to reduce communication overhead, and embedding adaptive federated modules to enhance long-term personalization. Additionally, exploring quantum-secure cryptographic primitives will strengthen the framework's resiliency and ensure its applicability in both current and post-quantum computing environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Thank you very much for your consideration.

Thank you very much for your consideration.

Data availability

No data was used for the research described in the article.

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